

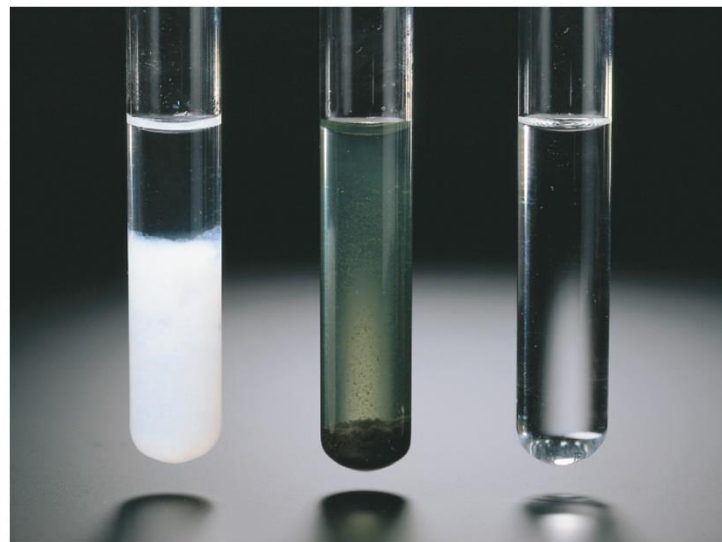
So you have a 30 minute quiz today

What are allowed?

- A pen
- A periodic table
- An A4-sized sheet of hand written notes
- A calculator

No other devices are allowed!

GENERAL CHEMISTRY I



Chapter 4 Reactions in Aqueous Solutions

Contents

- 4-1 The Nature of Aqueous Solutions
- 4-2 Precipitation Reactions
- 4-3 Acids, Bases, and Neutralization Reactions
- 4-4 Oxidation-Reduction Reactions
- 4-5 Concentrations of Solutions
- 4-6 Solution Stoichiometry and Chemical Analysis

4-1 The Nature of Aqueous Solutions

◆ Water (*Latin*: aqua)

- Inexpensive
- Can dissolve a vast number of substances
- Many substances dissociate into ions
- Aqueous solutions are found everywhere
 - Seawater
 - Living systems

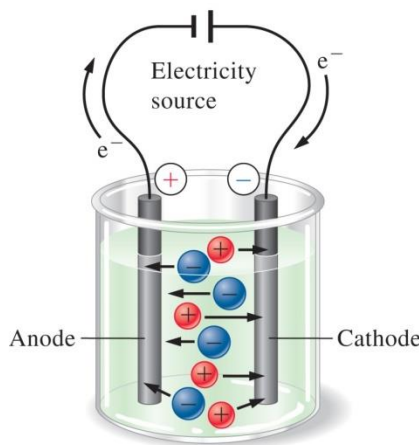


◆ A **solution** is a **homogeneous mixture** of two or more substances

- A **solvent** is the component of a solution that is present in the **greatest amount**
- A **solute** is the component of a solution that is **dissolved** in the solvent

Electrolytes and Nonelectrolytes

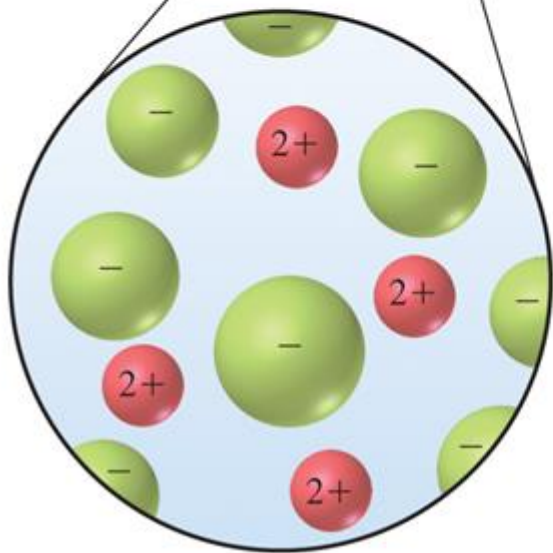
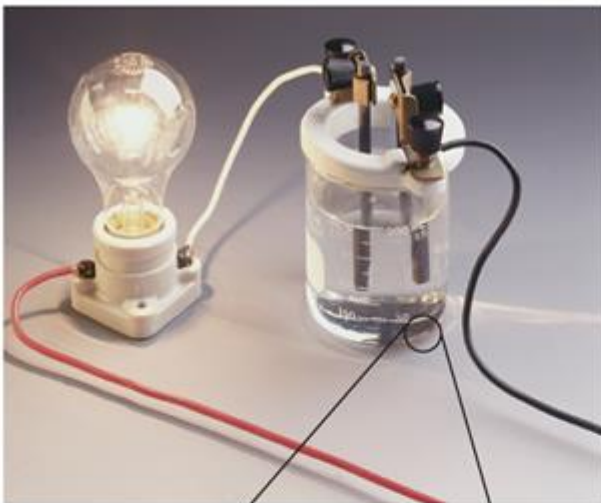
- ◆ Some solutes can dissolve in water and **dissociate** into ions.
- ➔ Electric charge can be carried.
- **Electrolyte**: A substance that **dissolves** in water and **separates** out into cations and anions



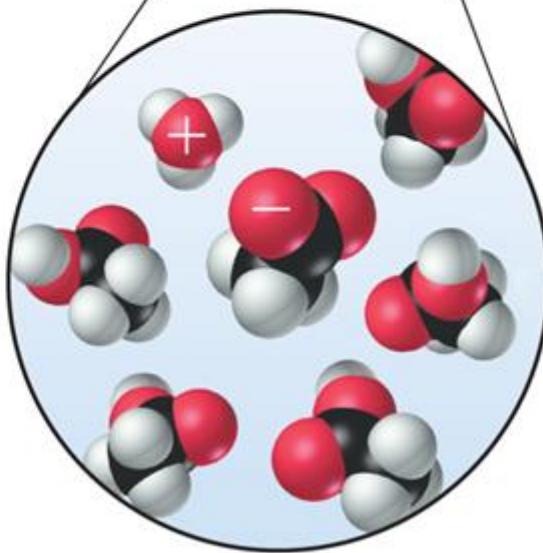
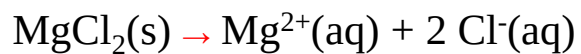
- ◆ Some solutes can dissolve in water but **cannot dissociate** into ions.
- ➔ Electric charge cannot be carried.
- **Nonelectrolyte**: A substance that **dissolves** in water and **does not separates** out into cations and anions

Types of Electrolytes

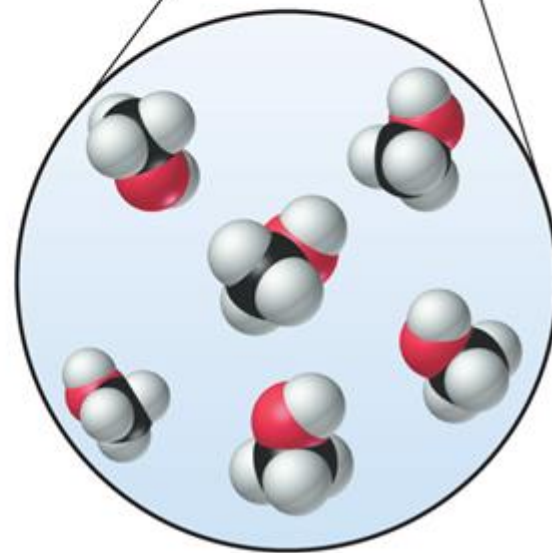
- ◆ *Strong electrolyte* completely dissociates into cation and anion .
 - Good electrical conduction.
- ◆ *Weak electrolyte* partially dissociates into cation and anion .
 - Fair conductor of electricity.
- ◆ *Non-electrolyte* does not dissociate.
 - Poor conductor of electricity.



A strong electrolyte



A weak electrolyte

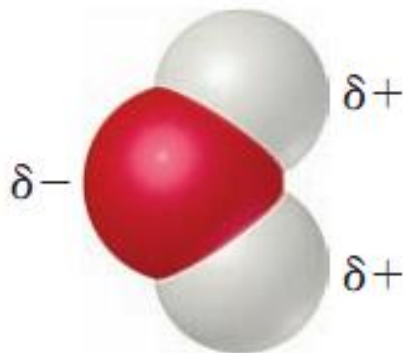


A non-electrolyte



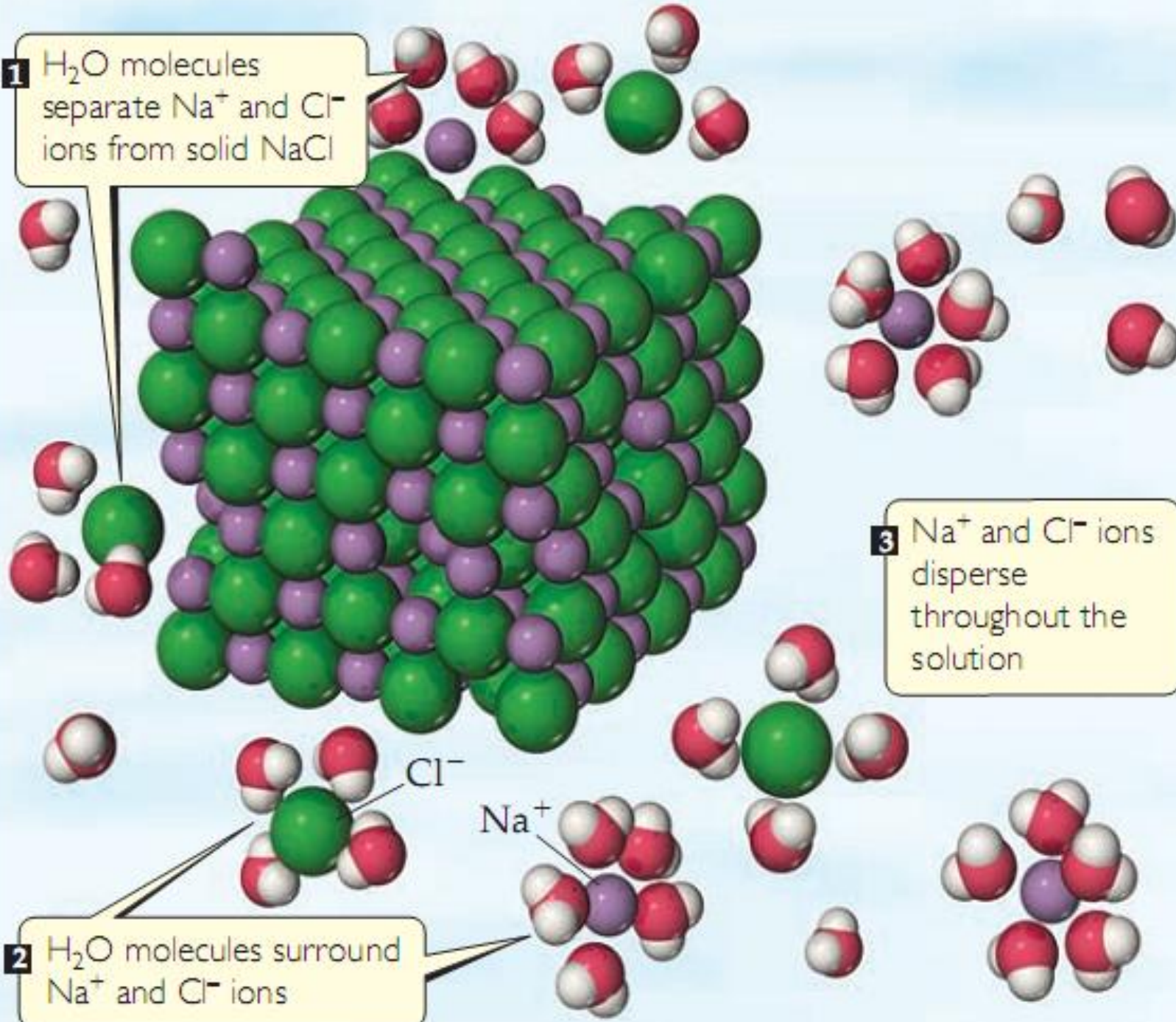


How Compounds Dissolve in Water



- the O atom is rich in electrons and has a partial negative charge
 - ➔ O atoms of water tend to stick to cations and bring them to the solution
- each H atom has a partial positive charge
 - ➔ H atoms of water tend to stick to anions and bring them to the solution

1 H_2O molecules separate Na^+ and Cl^- ions from solid NaCl



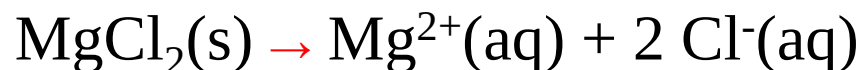
3 Na^+ and Cl^- ions disperse throughout the solution

2 H_2O molecules surround Na^+ and Cl^- ions

Dissociation and Ionization

An electrolyte produces ions in solution by one of two processes:

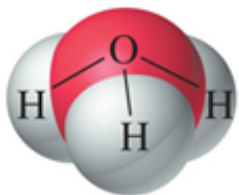
- **dissociation**, the separation of an entity into two or more entities
 - **Ionic** compounds
 - The ions separate from one another and become **surrounded** by water molecules



- **ionization**, the generation of one or more ions.
 - **Molecular** compounds
 - The ions are generated by a **reaction** of the compound with water

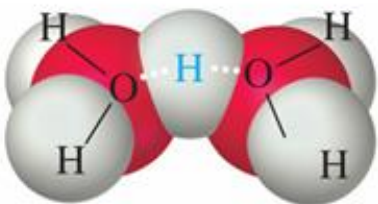


Solvation: The Hydrated Proton



Hydronium ion
 H_3O^+

The hydronium ion, interacts with other water molecules through electrostatic attractions.



A hydrated proton
 H_5O_2^+

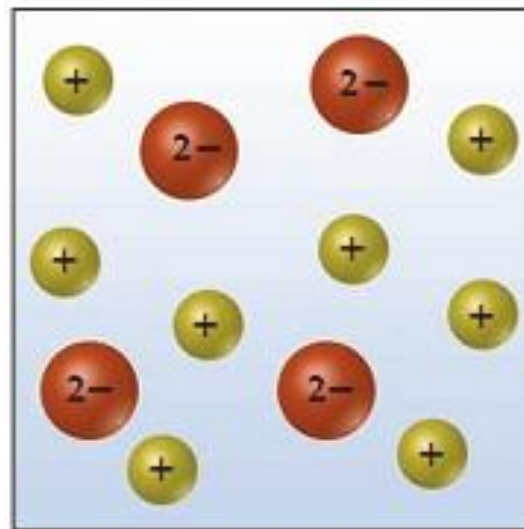


A hydrated proton
 H_9O_4^+

EXAMPLE

The accompanying diagram represents an aqueous solution of either MgCl_2 , KCl , or K_2SO_4 .

Which solution does the drawing best represent?



PRACTICE

1. If you have an aqueous solution that contains 1.5 moles of HCl, how many moles of ions are in the solution?

- (a) 1.0,
- (b) 1.5,
- (c) 2.0,
- (d) 2.5,
- (e) 3.0

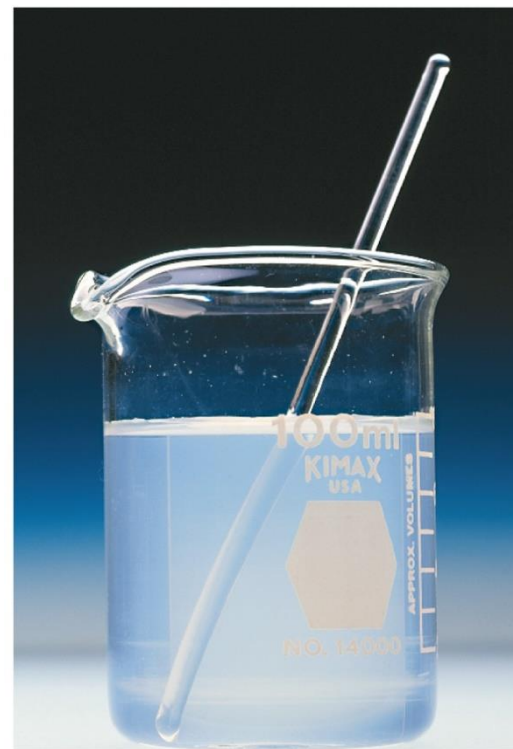
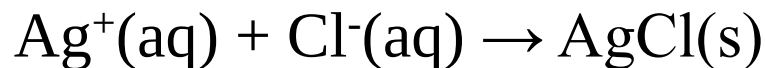
2. If you were to draw diagrams representing aqueous solutions of

- | | |
|------------------------------------|---------------|
| (a) NiSO_4 , | (a) 6 |
| (b) $\text{Ca}(\text{NO}_3)_2$, | (b) 12 |
| (c) Na_3PO_4 , | (c) 2 |
| (d) $\text{Al}_2(\text{SO}_4)_3$, | (d) 9 |

how many anions would you show if each diagram contained six cations?

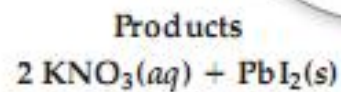
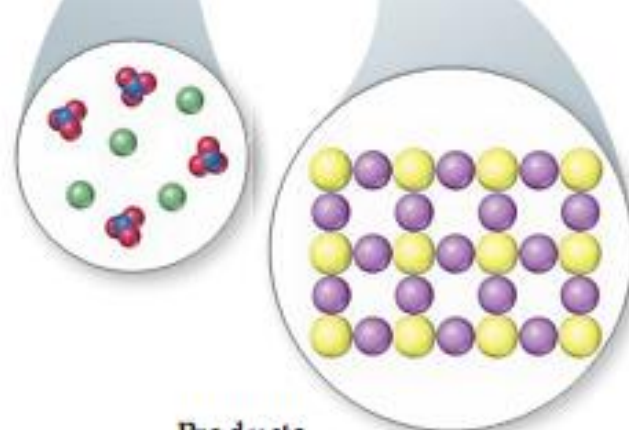
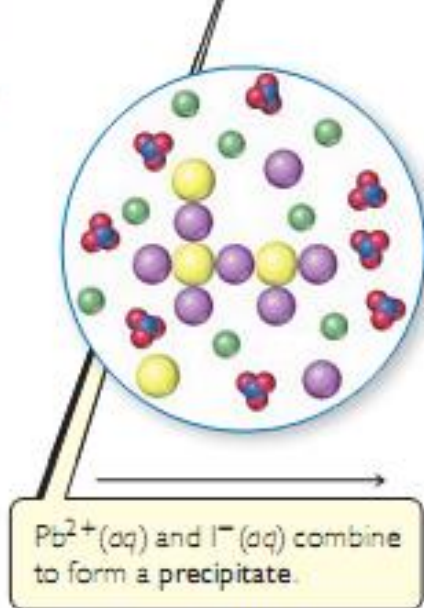
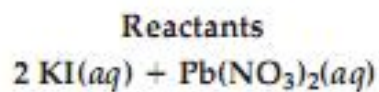
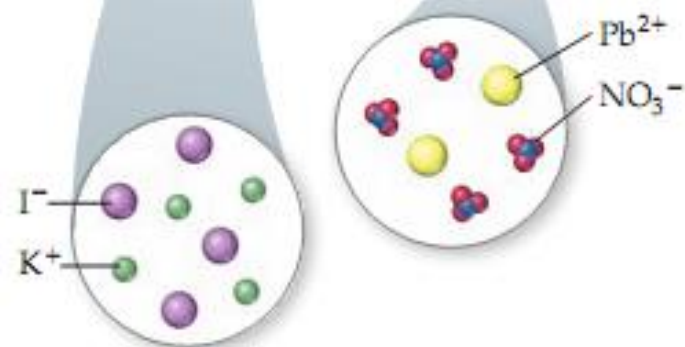
4-2 Precipitation Reactions

- ◆ Soluble ions can combine to form an *insoluble* compound.
- ◆ Precipitation occurs.
- ◆ A test for the presence of chloride ion in water.



Precipitation reactions transform ions into an insoluble salt in aqueous solution.

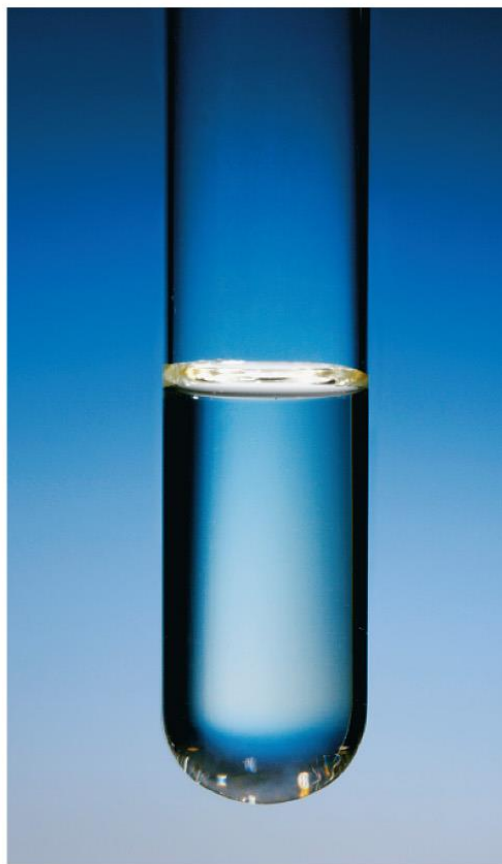
A **precipitate** is an insoluble solid formed by a reaction in solution.



Silver Nitrate and Sodium Iodide



$\text{AgNO}_3(\text{aq})$



$\text{NaI}(\text{aq})$



$\text{AgI}(\text{s})$



Exchange (Metathesis) Reactions

General equation:



Example:



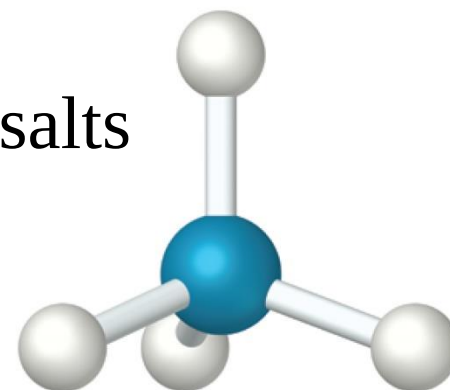
Complete and balance the equation for a metathesis reaction:

1. Use the chemical formulas of the reactants to determine which ions are present.
2. Write the chemical formulas of the products
3. Check the water solubilities of the products. For a precipitation reaction to occur, at least one product must be insoluble in water.
4. Balance the equation.

Solubility Rules

◆ Compounds that are *soluble*:

- Alkali metal ion and ammonium ion salts



Ammonium ion

- Nitrates, perchlorates and acetates



Solubility Rules

◆ Compounds that are *mostly soluble*:

- Chlorides, bromides and iodides Cl^- Br^- I^-

- Except those of Pb^{2+} , Ag^+ , and Hg_2^{2+} .

- Sulfates SO_4^{2-}

- Except those of Sr^{2+} , Ba^{2+} , Pb^{2+} and Hg_2^{2+} .

- $\text{Ca}(\text{SO}_4)$ is slightly soluble.

Solubility Rules

◆ Compounds that are *insoluble*:

- Hydroxides and sulfides



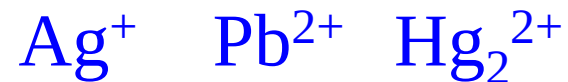
- Except alkali metal and ammonium salts
- Sulfides of alkaline earths are soluble
- Hydroxides of Sr^{2+} and Ca^{2+} are slightly soluble.

- Carbonates and phosphates



- Except alkali metal and ammonium salts

- Silver, Lead and Mercury



- Except nitrates, acetates and perchlorates

Solubility Guidelines for common ionic compounds in Water

Soluble Ionic Compounds		Important Exceptions
Compounds containing	NO_3^-	None
	CH_3COO^-	None
	Cl^-	Compounds of Ag^+ , Hg_2^{2+} , and Pb^{2+}
	Br^-	Compounds of Ag^+ , Hg_2^{2+} , and Pb^{2+}
	I^-	Compounds of Ag^+ , Hg_2^{2+} , and Pb^{2+}
	SO_4^{2-}	Compounds of Sr^{2+} , Ba^{2+} , Hg_2^{2+} , and Pb^{2+}
Insoluble Ionic Compounds		Important Exceptions
Compounds containing	S^{2-}	Compounds of NH_4^+ , the alkali metal cations, Ca^{2+} , Sr^{2+} , and Ba^{2+}
	CO_3^{2-}	Compounds of NH_4^+ and the alkali metal cations
	PO_4^{3-}	Compounds of NH_4^+ and the alkali metal cations
	OH^-	Compounds of NH_4^+ , the alkali metal cations, Ca^{2+} , Sr^{2+} , and Ba^{2+}

EXAMPLE

Classify these ionic compounds as soluble or insoluble in water:

(a) sodium carbonate, Na_2CO_3 ,

(a) Soluble

(b) lead sulfate, PbSO_4 .

(b) Insoluble

(a) Carbonates are insoluble except alkali metal salts

→ Na_2CO_3 is soluble

(b) Sulfates are mostly soluble except those of Pb^{2+}

→ PbSO_4 is insoluble

EXAMPLE

Given two separate aqueous solutions of BaCl_2 and K_2SO_4

(a) Predict the identity of the precipitate that forms when these are mixed.

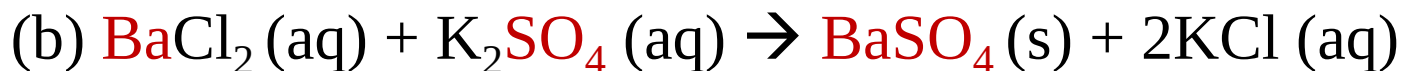
(b) Write the balanced chemical equation for the reaction.

Solve

(a) The reactants contain Ba^{2+} , Cl^- , K^+ , and SO_4^{2-} ions.

Exchanging the anions gives us BaSO_4 and KCl .

Most compounds of SO_4^{2-} are soluble but those of Ba^{2+} are not. Thus, BaSO_4 is insoluble and will precipitate from solution. KCl is soluble.



PRACTICE

1. Which of the following compounds is insoluble in water?

(a) $(\text{NH}_4)_2\text{S}$,

(a) s

(b) CaCO_3 ,

(b) insoluble

(c) NaOH ,

(c) s

(d) Ag_2SO_4 ,

(d) s

(e) $\text{Pb}(\text{CH}_3\text{COO})_2$.

(e) s

2. Classify the following compounds as soluble or insoluble in water:

(a) cobalt(II) hydroxide,

(a) insoluble

(b) barium nitrate,

(b) soluble

(c) ammonium phosphate.

(c) soluble

PRACTICE

1. Will a precipitate form when solutions of $\text{Ba}(\text{NO}_3)_2$ and KOH are mixed?

(a) Yes

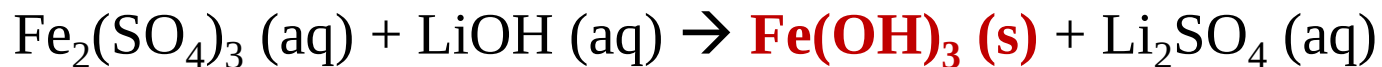
(b) No



2. When aqueous solutions of $\text{Fe}_2(\text{SO}_4)_3$ and LiOH are mixed

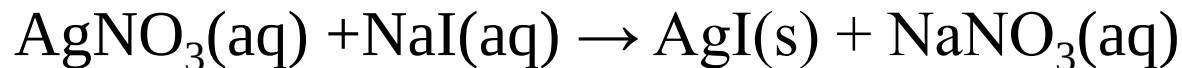
(a) What compound precipitates ?

(b) Write a balanced equation for the reaction.

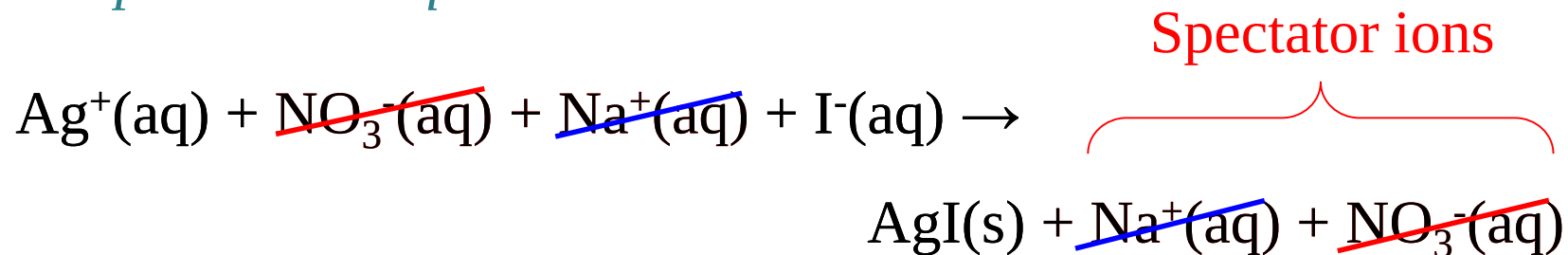


Net Ionic Equation

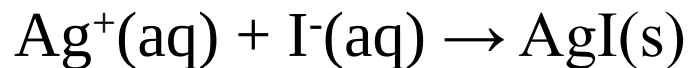
Overall Precipitation Reaction:



Complete ionic equation:



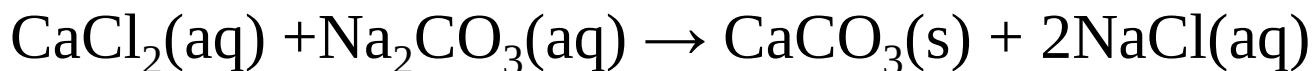
Net ionic equation:



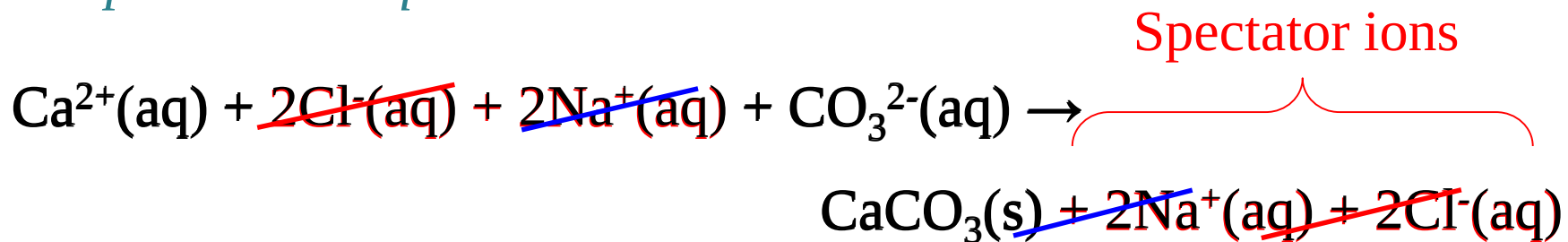
EXAMPLE

Write the net ionic equation for the precipitation reaction that occurs when aqueous solutions of calcium chloride and sodium carbonate are mixed.

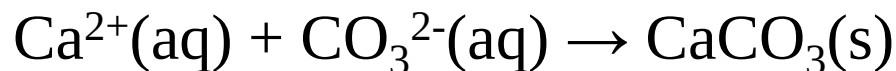
Overall Precipitation Reaction:



Complete ionic equation:



Net ionic equation:



PRACTICE

1. What happens when you mix an aqueous solution of sodium nitrate with an aqueous solution of barium chloride?

(a) There is no reaction; all possible products are soluble.

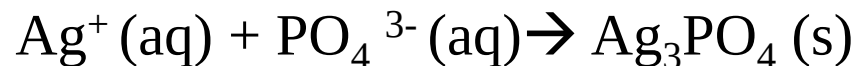
(b) Only barium nitrate precipitates.

(c) Only sodium chloride precipitates.

(d) Both barium nitrate and sodium chloride precipitate.

(e) Nothing; barium chloride is not soluble and it stays as a precipitate.

2. Write the net ionic equation for the precipitation reaction that occurs when aqueous solutions of silver nitrate and potassium phosphate are mixed.



4-3 Acid-Base Reactions

- ◆ Latin *acidus* (sour)
 - Sour taste
- ◆ Arabic *al-qali* (ashes of certain plants)
 - Bitter taste

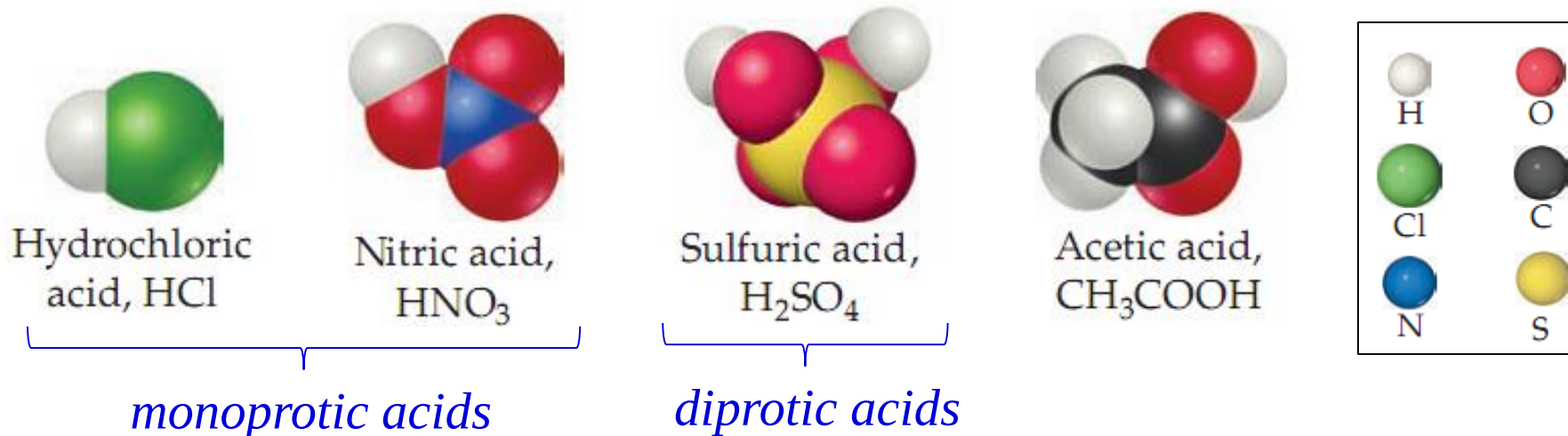


- ◆ Svante Arrhenius 1884 Acid-Base theory.

Acids are substances that dissociate in water to yield electrically charged atoms or molecules, called ions, one of which is a hydrogen ion (H^+).

Bases ionize in water to yield hydroxide ions (OH^-)

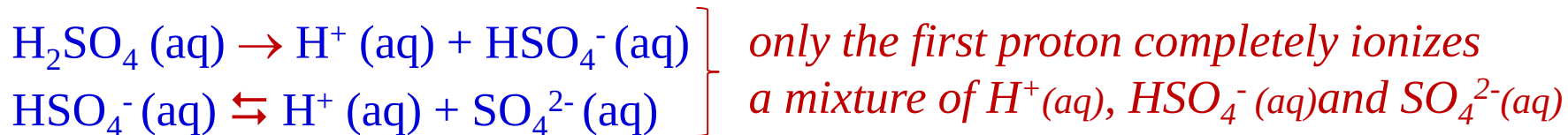
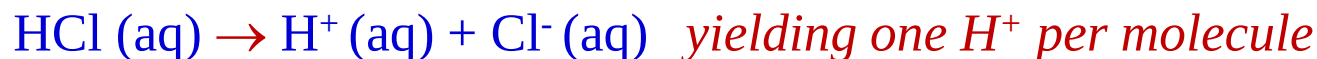
Acids



◆ Acids provide H⁺ in aqueous solution.

- Proton donor

◆ Strong acids completely ionize:

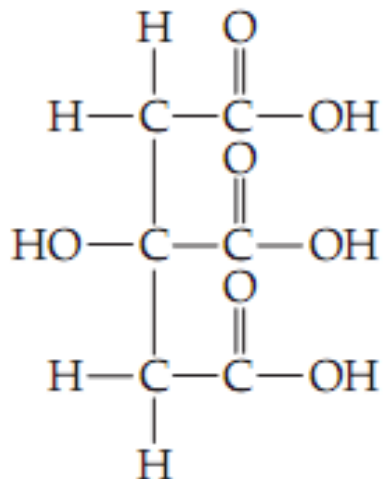


◆ Weak acid ionization is not complete:



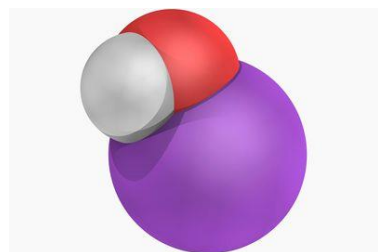
Give It Some Thought

The structural formula of citric acid, a main component of citrus fruits, is

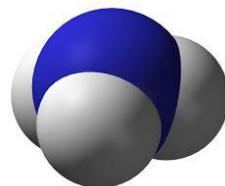


How many $\text{H}^+(\text{aq})$ can be generated by each citric acid molecule dissolved in water?

Bases



Sodium hydroxide,
NaOH



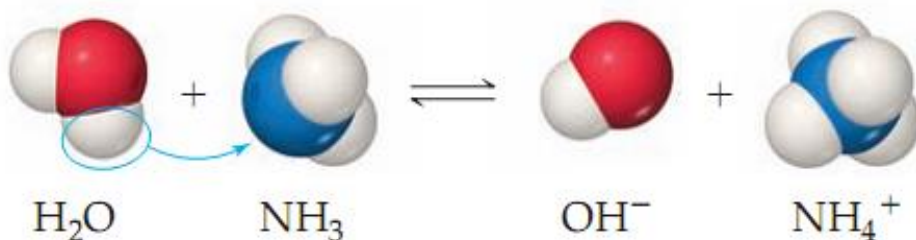
Ammonia,
NH₃

◆ Bases provide OH⁻ in aqueous solution.

◆ Strong bases:



◆ Weak bases:



*only about 1% of the NH₃ reacted
form a mixture of NH₃(aq), OH⁻(aq)
and NH₄⁺(aq)*

Recognizing Acids and Bases.

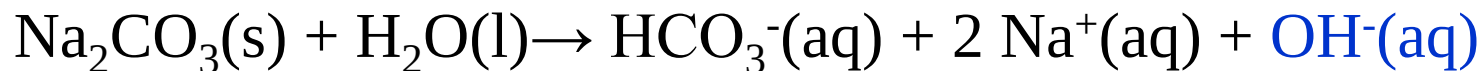
◆ Acids have ionizable hydrogen ions.



◆ Bases have OH^- combined with a metal ion.



or can be identified by chemical equations



Strong Acids

Hydrochloric acid, HCl

Hydrobromic acid, HBr

Hydroiodic acid, HI

Chloric acid, HClO_3

Perchloric acid, HClO_4

Nitric acid, HNO_3

Sulfuric acid (first proton), H_2SO_4

Strong Bases

Group 1A metal hydroxides

[LiOH, NaOH, KOH, RbOH, CsOH]

Heavy group 2A metal hydroxides

[$\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$]

Why isn't $\text{Al}(\text{OH})_3$ classified as a strong base?

a. $\text{Al}(\text{OH})_3$ is not basic in water.

b. $\text{Al}(\text{OH})_3$ is insoluble in water.

c. $\text{Al}(\text{OH})_3$ is a strong acid in water, not basic.

d. $\text{Al}(\text{OH})_3$ is a weak acid in water, not basic.

EXAMPLE

The following diagrams represent aqueous solutions of acids HX, HY, and HZ, with water molecules omitted for clarity. Rank the acids from strongest to weakest.

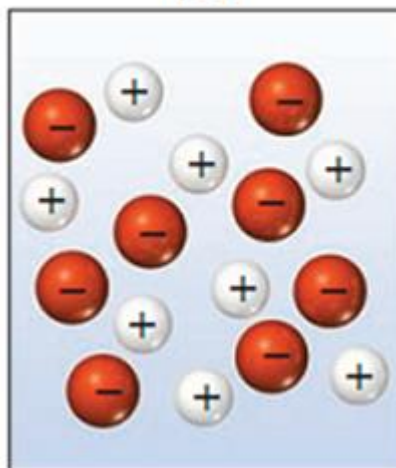
HX



*a mixture of molecules
and ions*

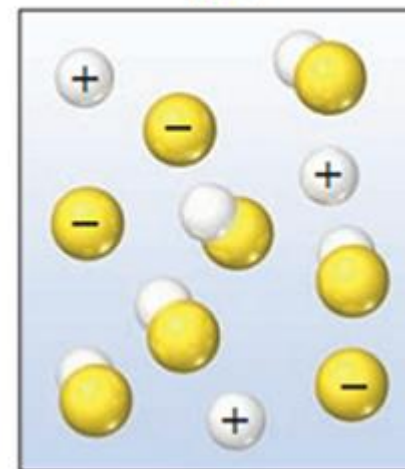
2 H⁺ ions 6 molecules

HY



totally ionized

HZ



*a mixture of molecules
and ions*

3 H⁺ ions 5 molecules

HY > HZ > HX

PRACTICE

A set of aqueous solutions are prepared containing different acids at the same concentration: acetic acid, chloric acid and hydrobromic acid. Which solution(s) are the most electrically conductive?

- (a) chloric acid,
- (b) hydrobromic acid,
- (c) acetic acid,
- (d)** both chloric acid and hydrobromic acid,
- (e) all three solutions have the same electrical conductivity

Identifying Strong and Weak Electrolytes

	Strong Electrolyte	Weak Electrolyte	Nonelectrolyte
Ionic	All	None	None
Molecular	Strong acids (see Table 4.2)	Weak acids, weak bases	All other compounds

Examples:

- ◆ CaCl_2 ➤ ionic ➤ strong electrolyte
- ◆ HNO_3 ➤ molecular, common strong acid ➤ strong electrolyte
- ◆ $\text{C}_2\text{H}_5\text{OH}$ (ethanol) ➤ molecular, neither an acid nor a base ➤ nonelectrolyte
- ◆ HCOOH (formic acid) ➤ molecular, weak acid ➤ weak electrolyte
- ◆ KOH ➤ ionic ➤ strong electrolyte

PRACTICE

1. Which of these substances, when dissolved in water, is a strong electrolyte?

- (a) ammonia,
- (b) hydrofluoric acid,
- (c) folic acid,
- (d) sodium nitrate,**
- (e) sucrose.

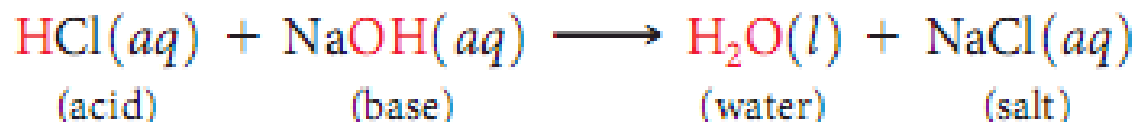
2. Consider solutions in which 0.1 mol of each of the following compounds is dissolved in 1 L of water: $\text{Ca}(\text{NO}_3)_2$ (calcium nitrate), $\text{C}_6\text{H}_{12}\text{O}_6$ (glucose), NaCH_3COO (sodium acetate), and CH_3COOH (acetic acid). Rank the solutions in order of increasing electrical conductivity, knowing that the greater the number of ions in solution, the greater the conductivity.



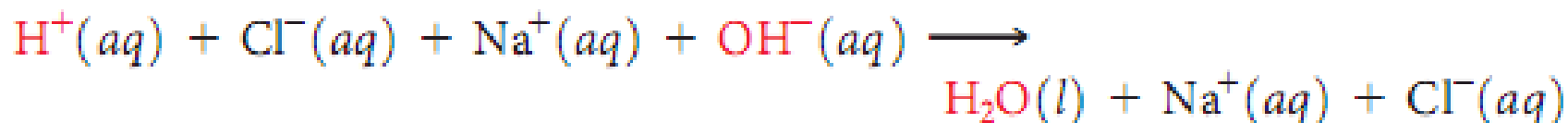
To be continued..

Neutralization Reactions and Salts

Molecular equation:



Complete ionic equation

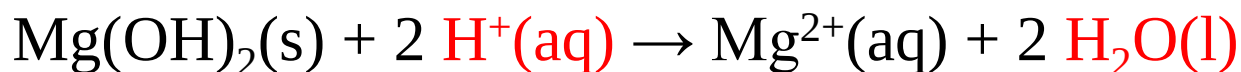


Net ionic equation



More Acid-Base Reactions

◆ Milk of magnesia



More Acid-Base Reactions

◆ Limestone and marble.

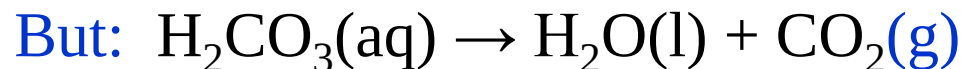
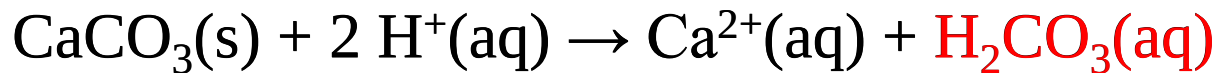


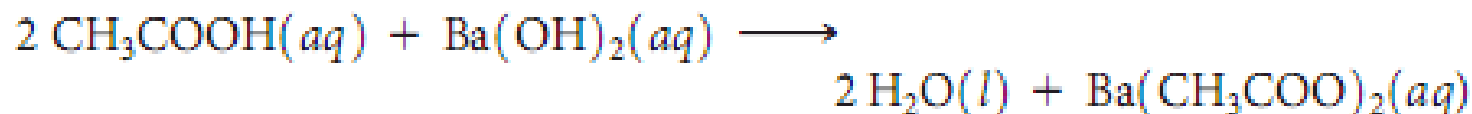
TABLE 5.3 Some Common Gas-Forming Reactions

Ion	Reaction
HSO_3^-	$\text{HSO}_3^- + \text{H}^+ \longrightarrow \text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
SO_3^{2-}	$\text{SO}_3^{2-} + 2 \text{H}^+ \longrightarrow \text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
HCO_3^-	$\text{HCO}_3^- + \text{H}^+ \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
CO_3^{2-}	$\text{CO}_3^{2-} + 2 \text{H}^+ \longrightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
S^{2-}	$\text{S}^{2-} + 2 \text{H}^+ \longrightarrow \text{H}_2\text{S}(\text{g})$
NH_4^+	$\text{NH}_4^+ + \text{OH}^- \longrightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$

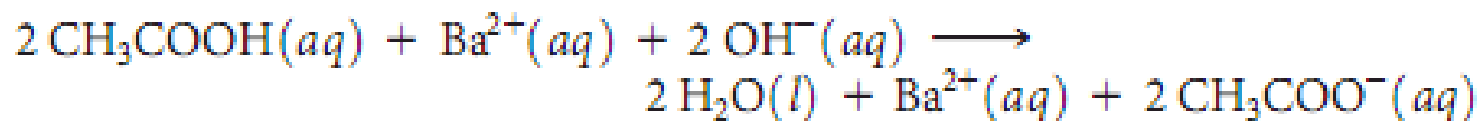
EXAMPLE

For the reaction between aqueous solutions of acetic acid (CH_3COOH) and barium hydroxide, $\text{Ba}(\text{OH})_2$, write

(a) the balanced molecular equation,



(b) the complete ionic equation,

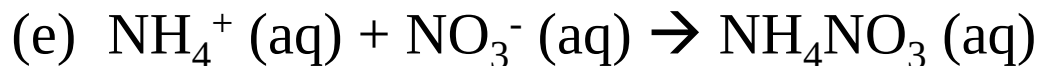
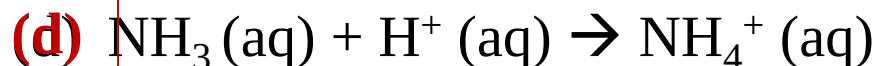
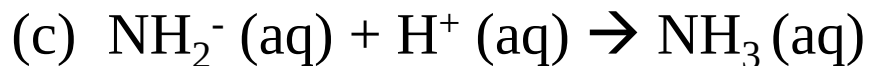
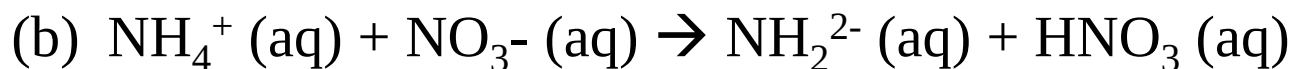
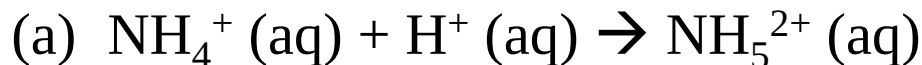


(c) the net ionic equation.



PRACTICE

1. Which is the correct net ionic equation for the reaction of aqueous ammonia with nitric acid?



2. For the reaction of phosphorous acid (H_3PO_3) and potassium hydroxide (KOH), write

(a) the balanced molecular equation



(b) the net ionic equation



Last week

- 4-1 The Nature of Aqueous Solutions
- 4-2 Precipitation Reactions
- 4-3 Acids, Bases, and Neutralization Reactions

This week

- 4-4 Oxidation-Reduction Reactions
- 4-5 Concentrations of Solutions
- 4-6 Solution Stoichiometry and Chemical Analysis

4-4 Oxidation-Reduction (Redox) Reactions

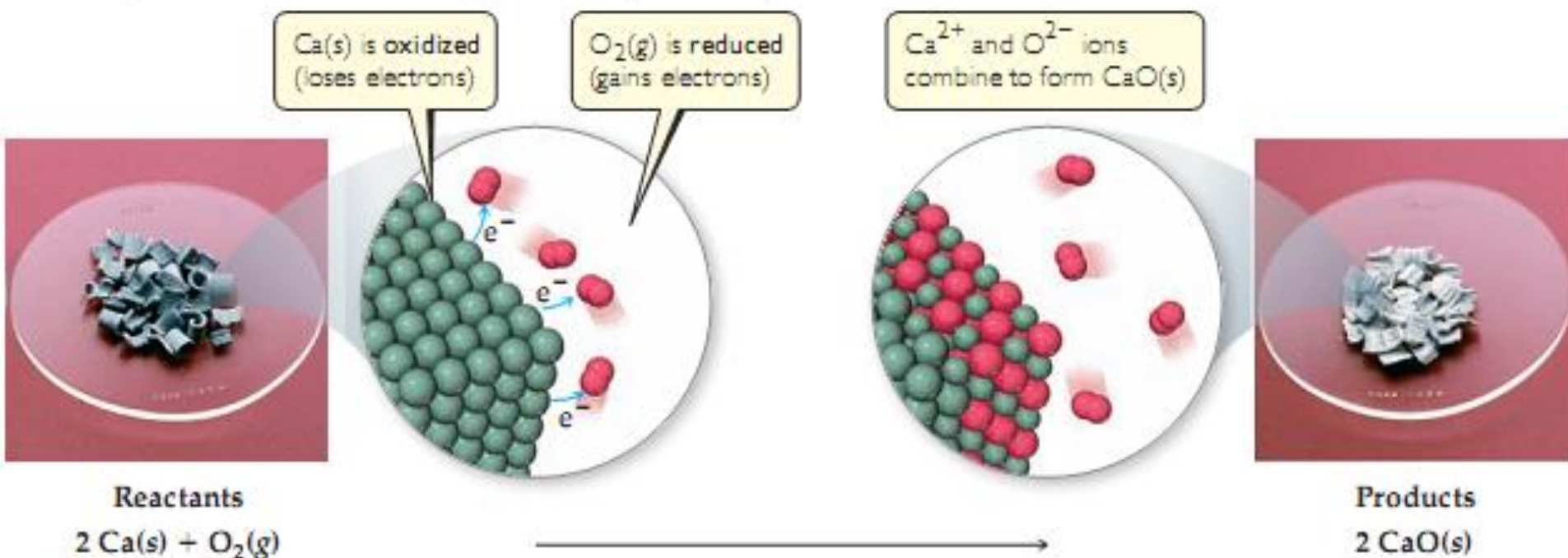
In **precipitation** reactions, **cations and anions** come together to form an **insoluble** ionic compound.

In **neutralization** reactions, **protons are transferred** from one reactant to another.

In **redox** reactions, **electrons are transferred** from one reactant to another

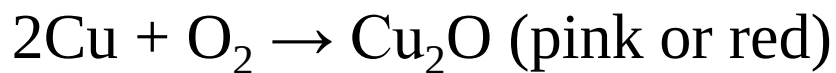
- Loss of electrons by a substance is called oxidation.
- The gain of electrons by a substance is called reduction
- Oxidation and reduction always occur together.

How many electrons does each oxygen atom gain during the course of this reaction?

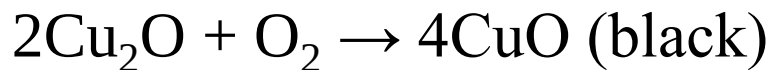




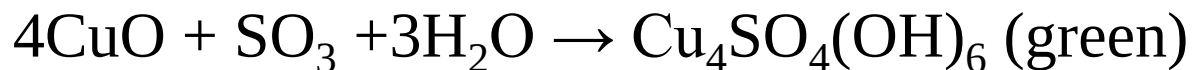
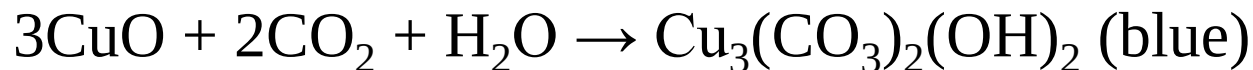
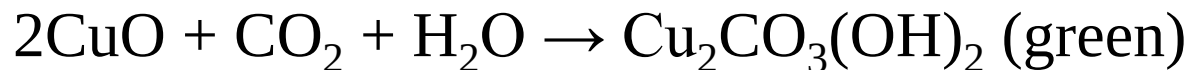
Initially,

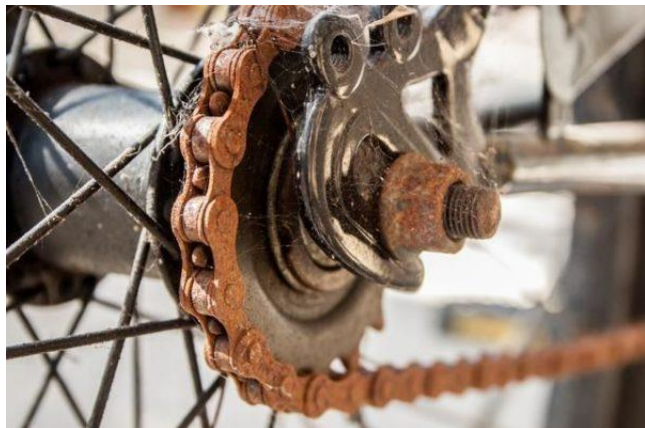


Then



Due to pollution from coal firing:

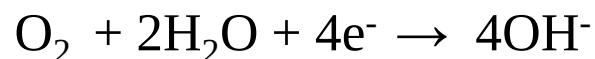




Rust consists of

- hydrated iron(III) oxides $\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$, and
- iron(III) oxide-hydroxide ($\text{FeO}(\text{OH})$, $\text{Fe}(\text{OH})_3$).

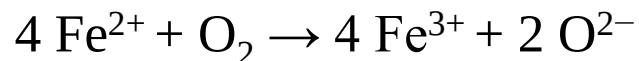
- Reduction of oxygen that is dissolved into water :



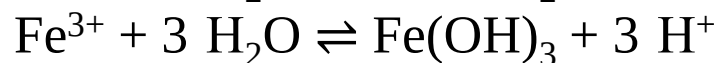
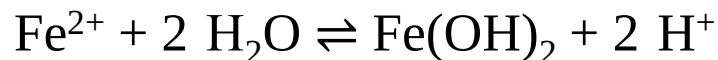
- Oxidation of iron in the presence of water:



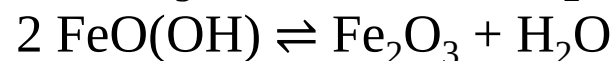
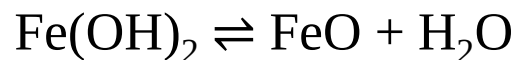
- Rust is formed in the presence of water via a redox reaction:



- Multistep acid–base reactions



- Dehydration equilibria



Oxidation Numbers

1. For an atom in its **elemental form**, the oxidation number is always **zero**.
2. For any **monatomic ion** the oxidation number equals the **ionic charge**.
3. **Nonmetals** usually have **negative** oxidation numbers, although they can sometimes be positive:
 - (a) O usually has an oxidation number of -2 in both ionic and molecular compounds, except for peroxides, which contain the O_2^{2-} ion, giving each oxygen an oxidation number of -1.
 - (b) H usually has an oxidation number of +1 when bonded to nonmetals and -1 when bonded to metals (for example, metal hydrides such as sodium hydride, NaH).
 - (c) F has an oxidation number of -1 in all compounds. The other halogens have an oxidation number of -1 in most binary compounds. When combined with oxygen, as in oxyanions, they have positive oxidation states.
4. The sum of the oxidation numbers of all atoms in a neutral compound is zero, in a polyatomic ion equals the charge of the ion.

Oxidation and Reduction

◆ Oxidation

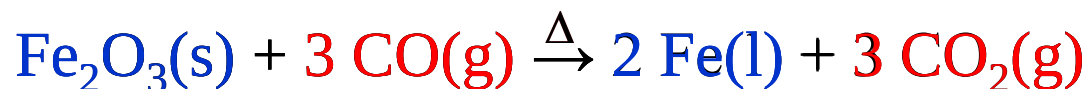
- Oxidation number of some element *increases* in the reaction.
- Electrons are on the right of the equation

◆ Reduction

- Oxidation number of some element *decreases* in the reaction.
- Electrons are on the left of the equation.

Some General Principles

- ◆ Example: Hematite is converted to iron in a blast furnace.

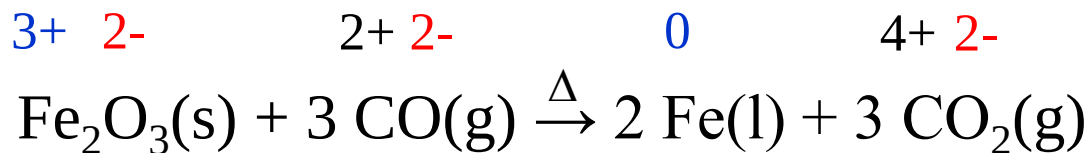


- ◆ Oxidation and reduction always occur together.

Fe^{3+} is reduced to metallic iron.

$\text{CO}(\text{g})$ is oxidized to carbon dioxide.

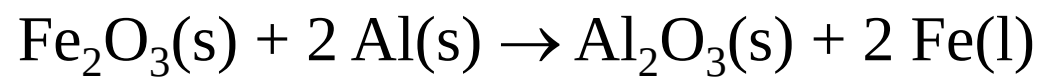
- ◆ Assign oxidation number:



Fe^{3+} is reduced to metallic iron.

$\text{CO}(\text{g})$ is oxidized to carbon dioxide.

Thermite Reaction



An Oxidation Reduction Reaction



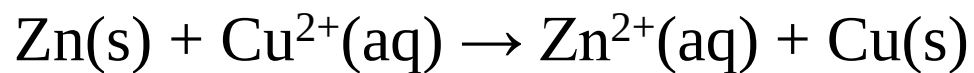
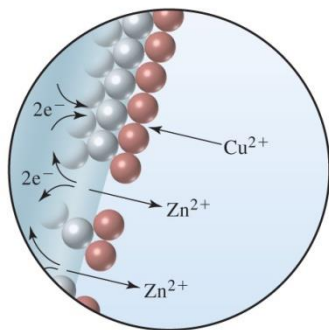
(a)



(b)



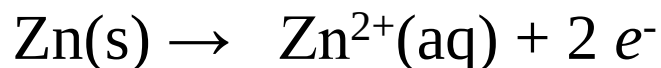
(c)



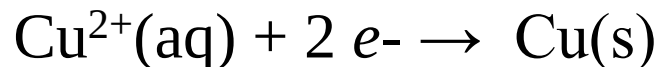
Oxidation and Reduction Half-Reactions

- ◆ A reaction represented by two half-reactions.

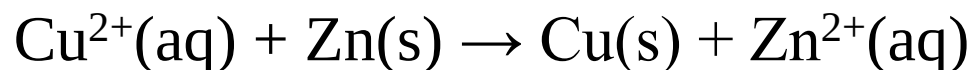
Oxidation:



Reduction:



Overall:



EXAMPLE

Determine the oxidation number of sulfur in

- (a) H_2S , (a) When bonded to a nonmetal, hydrogen has an oxidation number of +1. Because the H_2S molecule is neutral, the sum of the oxidation numbers must equal zero. Letting x equal the oxidation number of S, we have $2(+1) + x = 0$. Thus, **S has an oxidation number of -2.**
- (b) S_8 , (b) Because S_8 is an elemental form of sulfur, **the oxidation number of S is 0.**
- (c) SCl_2 , (c) Because SCl_2 is a binary compound, we expect chlorine to have an oxidation number of -1. The sum of the oxidation numbers must equal zero. Letting x equal the oxidation number of S, we have $x + 2(-1) = 0$. Thus, **the oxidation number of S must be +2.**
- (d) Na_2SO_3 , (d) Sodium, an alkali metal, always has an oxidation number of +1 in its compounds. Oxygen commonly has an oxidation state of -2. Letting x equal the oxidation number of S, we have $2(+1) + x + 3(-2) = 0$. Therefore, the **oxidation number of S in Na_2SO_3 is +4.**
- (e) SO_4^{2-} , (e) The oxidation state of O is -2. The sum of the oxidation numbers equals -2, the net charge of the SO_4^{2-} ion. Thus, we have $x + 4(-2) = -2$. Therefore the **oxidation number of S in this ion is +6.**

PRACTICE

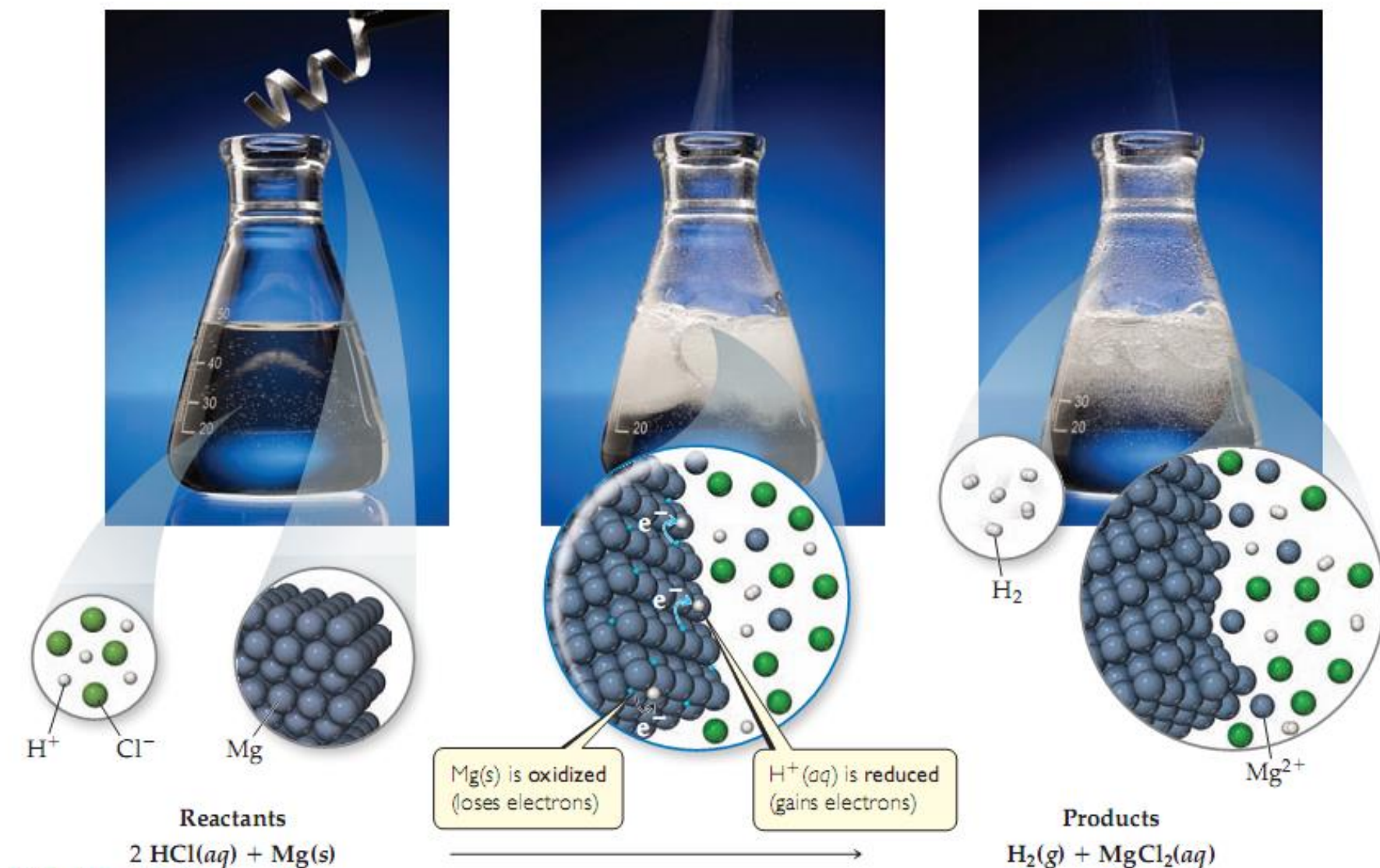
1. In which compound is the oxidation state of oxygen -1?

- (a) O_2 ,
- (b) H_2O ,
- (c) H_2SO_4 ,
- (d) H_2O_2 ,
- (e) KCH_3COO

2. What is the oxidation state of the boldfaced element in

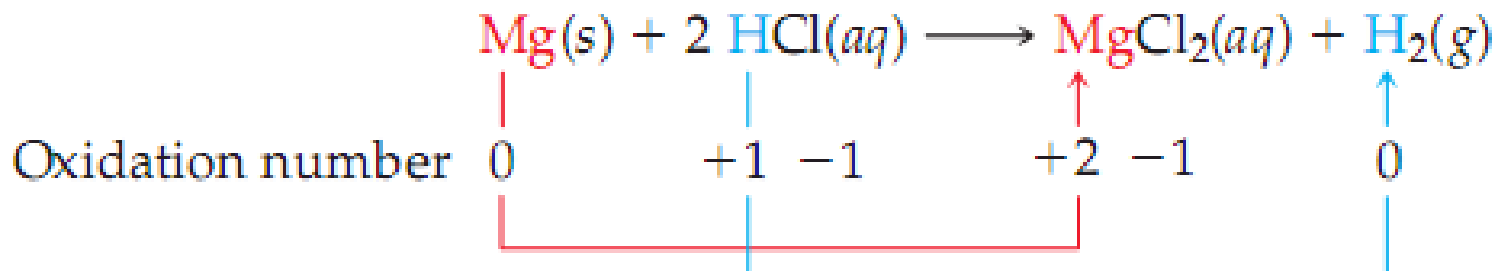
- (a) **P** $_2\text{O}_5$,
- (b) Na **H**,
- (c) **Cr** $_2\text{O}_7^{2-}$,
- (d) **Sn** Br_4 ,
- (e) Ba **O** $_2$?

How many moles of hydrogen gas would be produced for every mole of magnesium added into the HCl solution?



Oxidation of Metals by Acids and Salts

- ◆ Few can be balanced by inspection.
- ◆ Systematic approach required.
- ◆ The Half-Reaction (Ion-Electron) Method



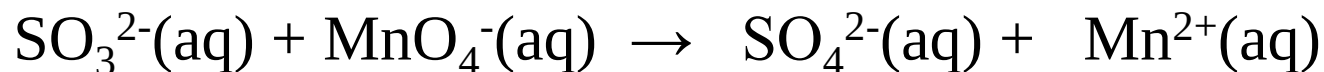
Balancing in Acid

- ◆ Write the equations for the half-reactions.
 - Balance all atoms except H and O.
 - Balance oxygen using H_2O .
 - Balance hydrogen using H^+ .
 - Balance charge using e^- .
- ◆ Equalize the number of electrons.
- ◆ Add the half reactions.
- ◆ Check the balance.

EXAMPLE

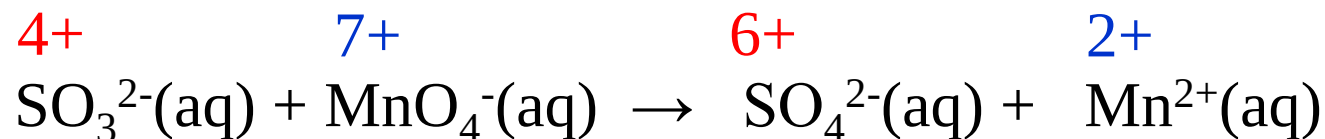
Balancing the Equation for a Redox Reaction in Acidic

Solution. The reaction described below is used to determine the sulfite ion concentration present in wastewater from a papermaking plant. Write the balanced equation for this reaction in acidic solution.

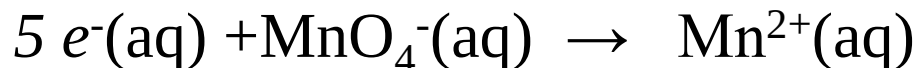


EXAMPLE

Determine the oxidation states:



Write the half-reactions:

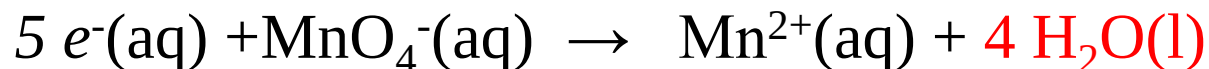
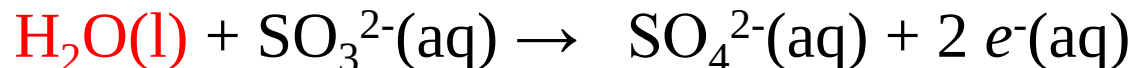


Balance atoms other than H and O:

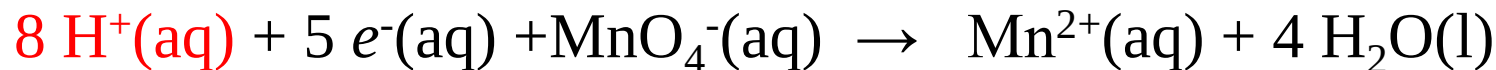
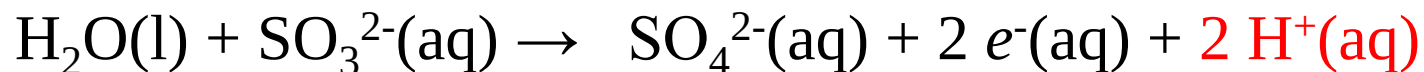
Already balanced for elements.

EXAMPLE

Balance O by adding H₂O:



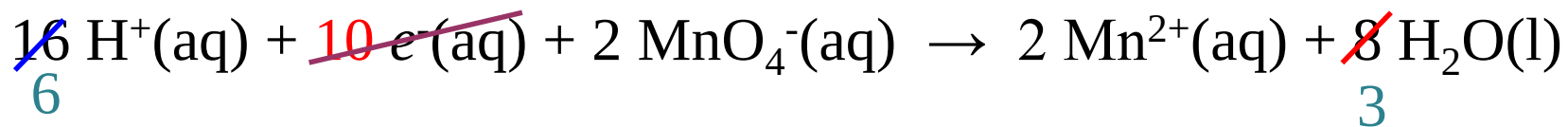
Balance hydrogen by adding H⁺:



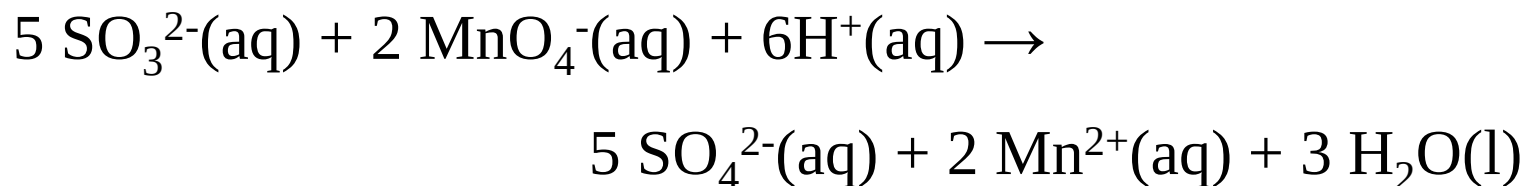
Check that the charges are balanced: Add e⁻ if necessary.

EXAMPLE

Multiply the half-reactions to balance all e^- :



Add both equations and simplify:



Check the balance!

PRACTICE

1. Which of the following statements is true about the reaction between zinc and copper sulfate?

- (a) Zinc is oxidized, and copper ion is reduced.
- (b) Zinc is reduced, and copper ion is oxidized.
- (c) All reactants and products are soluble strong electrolytes.
- (d) The oxidation state of copper in copper sulfate is 0.
- (e) More than one of the previous choices are true.

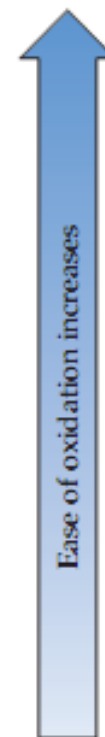
2. For the reaction between magnesium and cobalt(II) sulfate

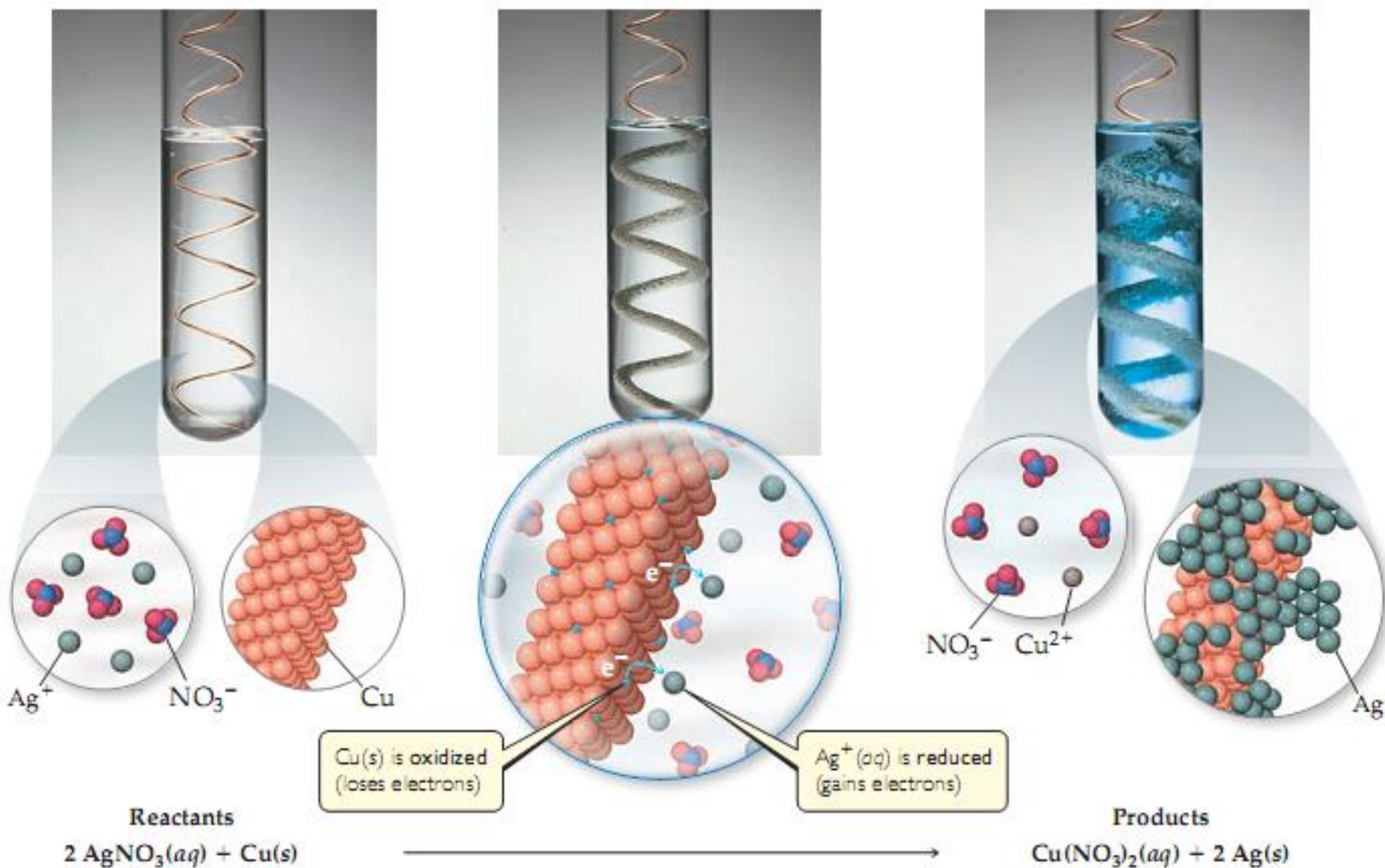
- (a) Write the balanced molecular and net ionic equations .
- (b) What is oxidized and what is reduced in the reaction?

The Activity Series

Any metal on the list can be oxidized by the ions of elements below it

Metal	Oxidation Reaction
Lithium	$\text{Li}(s) \longrightarrow \text{Li}^+(aq) + e^-$
Potassium	$\text{K}(s) \longrightarrow \text{K}^+(aq) + e^-$
Barium	$\text{Ba}(s) \longrightarrow \text{Ba}^{2+}(aq) + 2e^-$
Calcium	$\text{Ca}(s) \longrightarrow \text{Ca}^{2+}(aq) + 2e^-$
Sodium	$\text{Na}(s) \longrightarrow \text{Na}^+(aq) + e^-$
Magnesium	$\text{Mg}(s) \longrightarrow \text{Mg}^{2+}(aq) + 2e^-$
Aluminum	$\text{Al}(s) \longrightarrow \text{Al}^{3+}(aq) + 3e^-$
Manganese	$\text{Mn}(s) \longrightarrow \text{Mn}^{2+}(aq) + 2e^-$
Zinc	$\text{Zn}(s) \longrightarrow \text{Zn}^{2+}(aq) + 2e^-$
Chromium	$\text{Cr}(s) \longrightarrow \text{Cr}^{3+}(aq) + 3e^-$
Iron	$\text{Fe}(s) \longrightarrow \text{Fe}^{2+}(aq) + 2e^-$
Cobalt	$\text{Co}(s) \longrightarrow \text{Co}^{2+}(aq) + 2e^-$
Nickel	$\text{Ni}(s) \longrightarrow \text{Ni}^{2+}(aq) + 2e^-$
Tin	$\text{Sn}(s) \longrightarrow \text{Sn}^{2+}(aq) + 2e^-$
Lead	$\text{Pb}(s) \longrightarrow \text{Pb}^{2+}(aq) + 2e^-$
Hydrogen	$\text{H}_2(g) \longrightarrow 2\text{H}^+(aq) + 2e^-$
Copper	$\text{Cu}(s) \longrightarrow \text{Cu}^{2+}(aq) + 2e^-$
Silver	$\text{Ag}(s) \longrightarrow \text{Ag}^+(aq) + e^-$
Mercury	$\text{Hg}(l) \longrightarrow \text{Hg}^{2+}(aq) + 2e^-$
Platinum	$\text{Pt}(s) \longrightarrow \text{Pt}^{2+}(aq) + 2e^-$
Gold	$\text{Au}(s) \longrightarrow \text{Au}^{3+}(aq) + 3e^-$





Give It Some Thought

Does a reaction occur when an aqueous solution of NiCl_2 (aq)

(a) is added to a test tube containing strips of metallic zinc,

(b) is added to a test tube containing $\text{Zn}(\text{NO}_3)_2$ (aq)?

EXAMPLE

Will an aqueous solution of iron(II) chloride oxidize magnesium metal? If so, write the balanced molecular and net ionic equations for the reaction.

Because Mg is above Fe in the table, the reaction occurs



Both FeCl_2 and MgCl_2 are soluble strong electrolytes and can be written in ionic form



PRACTICE

1. Which of these metals is the easiest to oxidize?

- (a) gold,
- (b) lithium,
- (c) iron,
- (d) sodium,
- (e) aluminum.

2. Which of the following metals will be oxidized by $\text{Pb}(\text{NO}_3)_2$?

- (a) Zn,
- (b) Cu,
- (c) Fe

Balancing in Basic Solution

- ◆ OH^- appears instead of H^+ .
- ◆ Treat the equation as if it were in acid.
 - Then add OH^- to each side to neutralize H^+ .
 - Remove H_2O appearing on both sides of equation.
- ◆ Check the balance.

Disproportionation Reactions

- ◆ The same substance is both oxidized and reduced.
- ◆ Some have practical significance

- Hydrogen peroxide



- Sodium thiosulphate



Oxidizing and Reducing Agents.

- ◆ An oxidizing agent (**oxidant**):
 - Contains an element whose oxidation state *decreases* in a redox reaction
- ◆ A reducing agent (**reductant**):
 - Contains an element whose oxidation state *increases* in a redox reaction.

Oxidation States of Nitrogen

Compound or ion	Oxidation state
NO_3^-	+5
N_2O_4	+4
NO_2^-	+3
NO	+2
N_2O	+1
N_2	0
NH_2OH	-1
N_2H_4	-2
NH_3	-3

This species cannot be oxidized further

This species cannot be reduced further

Oxidation half-reaction (reducing agent)

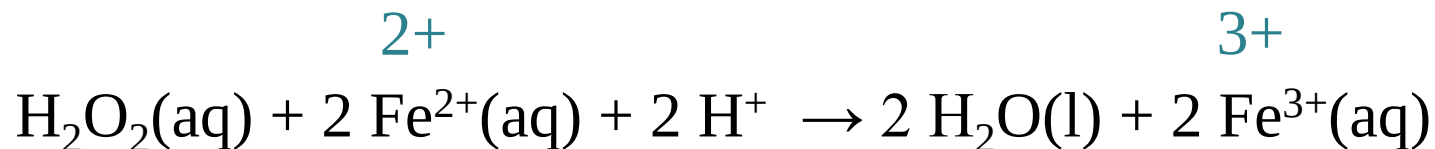
Reduction half-reaction (oxidizing agent)

EXAMPLE

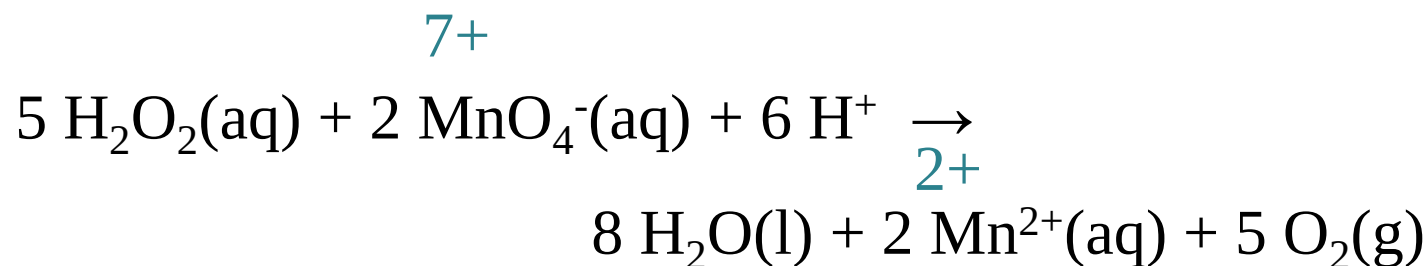
Identifying Oxidizing and Reducing Agents. Hydrogen peroxide, H_2O_2 , is a versatile chemical. Its uses include bleaching wood pulp and fabrics and substituting for chlorine in water purification. One reason for its versatility is that it can be either an oxidizing or a reducing agent. For the following reactions, identify whether hydrogen peroxide is an oxidizing or reducing agent.



EXAMPLE



Iron is oxidized and
peroxide is reduced.



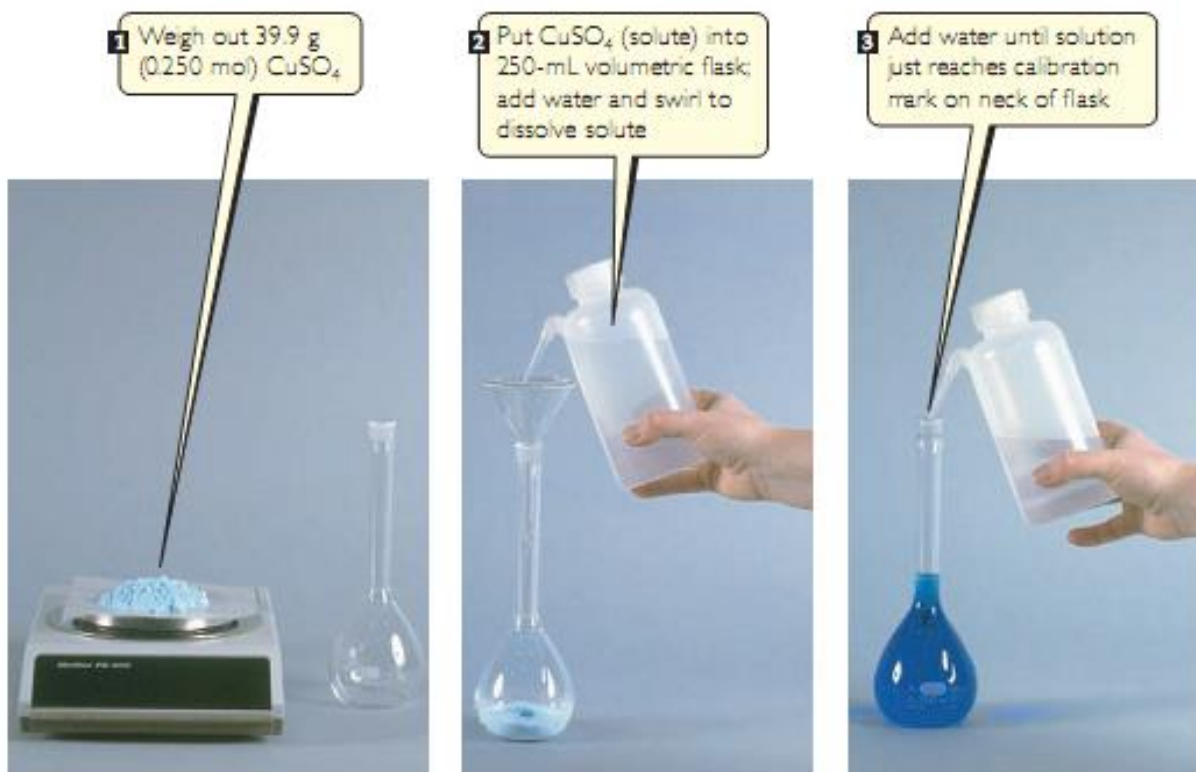
Manganese is reduced and
peroxide is oxidized.

4-5 Concentrations of Solutions.

◆ Molarity

$$\text{Molarity} = \frac{\text{moles solute}}{\text{volume of solution in liters}}$$

Example: Preparing 0.250 L of a 1.00 M solution of CuSO_4 .



EXAMPLE

Calculate the molarity of a solution made by dissolving 23.4 g of sodium sulfate in enough water to form 125 mL of solution.

$$\text{Moles Na}_2\text{SO}_4 = (23.4 \text{ g Na}_2\text{SO}_4) \left(\frac{1 \text{ mol Na}_2\text{SO}_4}{142.1 \text{ g Na}_2\text{SO}_4} \right) = 0.165 \text{ mol Na}_2\text{SO}_4$$

$$\text{Liters soln} = (125 \text{ mL}) \left(\frac{1 \text{ L}}{1000 \text{ mL}} \right) = 0.125 \text{ L}$$

$$\text{Molarity} = \frac{0.165 \text{ mol Na}_2\text{SO}_4}{0.125 \text{ L soln}} = 1.32 \frac{\text{mol Na}_2\text{SO}_4}{\text{L soln}} = 1.32 \text{ M}$$

PRACTICE

1. What is the molarity of a solution that is made by dissolving 3.68 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in sufficient water to form 275.0 mL of solution?

- (a) 13.4 M,
- (b) 7.43×10^{-2} M,
- (c) 3.91×10^{-2} M
- (d) 7.43×10^{-5} M
- (e) 3.91×10^{-5} M.

2. Calculate the molarity of a solution made by dissolving 5.00 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in sufficient water to form exactly 100 mL of solution.

Interconverting Molarity and Volume

If we know the molarity of an HNO_3 solution to be 0.200 M, we can calculate the number of moles of solute in a given volume, say 2.0 L.

$$\text{Moles HNO}_3 = (2.0 \text{ L soln}) \left(\frac{0.200 \text{ mol HNO}_3}{1 \text{ L soln}} \right) = 0.40 \text{ mol HNO}_3$$

$$\text{Liters soln} = (2.0 \text{ mol HNO}_3) \left[\frac{1 \text{ L soln}}{0.30 \text{ mol HNO}_3} \right] = 6.7 \text{ L soln}$$

$$\text{Liters} = \text{moles} \times 1/M = \cancel{\text{moles}} \times \text{liters}/\cancel{\text{mole}}.$$

Interconverting Percent Concentration and Molarity

A typical American beer contains 5.0% ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) by volume in water (along with other components). The density of ethanol is 0.789 g/mL. What is the molarity of ethanol?

We would first consider 1.00 L of beer

$$5\% = 5/100 = 0.05 \text{ L}$$

$$\text{Moles ethanol} = (0.050 \text{ L}) \left(\frac{1000 \text{ mL}}{\text{L}} \right) \left(\frac{0.789 \text{ g}}{\text{mL}} \right) \left(\frac{1 \text{ mol}}{46.0 \text{ g}} \right) = 0.858 \text{ mol}$$

There are 0.858 moles of ethanol in 1.00 L of beer, the concentration of ethanol in beer is **0.86 M**

Dilution

How we can prepare a 0.100 M CuSO_4 solution from 1.00 M CuSO_4

Moles solute before dilution = moles solute after dilution

$$\begin{aligned}\text{Moles CuSO}_4 \text{ in dilute soln} &= (0.2500 \cancel{\text{ L soln}}) \left(\frac{0.100 \text{ mol CuSO}_4}{\cancel{\text{ L soln}}} \right) \\ &= 0.0250 \text{ mol CuSO}_4\end{aligned}$$

$$\text{Liters of conc soln} = (0.0250 \cancel{\text{ mol CuSO}_4}) \left(\frac{1 \text{ L soln}}{1.00 \cancel{\text{ mol CuSO}_4}} \right) = 0.0250 \text{ L}$$

1 Draw 25.0 mL of 1.00 M stock solution into pipette



2 Add concentrated solution in pipette to 250-mL volumetric flask



3 Dilute with water until solution reaches calibration mark on neck of flask and mix to create 0.100 M solution



EXAMPLE

How many milliliters of 3.0 M H_2SO_4 are needed to make 450 mL of 0.10 M H_2SO_4 ?

$$\begin{aligned}\text{Moles } \text{H}_2\text{SO}_4 \text{ in dilute solution} &= (0.450 \text{ L soln}) \left(\frac{0.10 \text{ mol } \text{H}_2\text{SO}_4}{1 \text{ L soln}} \right) \\ &= 0.045 \text{ mol } \text{H}_2\text{SO}_4\end{aligned}$$

$$\text{L conc soln} = (0.045 \text{ mol } \text{H}_2\text{SO}_4) \left(\frac{1 \text{ L soln}}{3.0 \text{ mol } \text{H}_2\text{SO}_4} \right) = 0.015 \text{ L soln}$$

Converting liters to milliliters gives 15 mL

PRACTICE

1. What volume of a 1.00 M stock solution of glucose must be used to make 500.0 mL of a 1.75×10^{-2} M glucose solution in water?

(a) 1.75 mL, (b) 8.75 mL, (c) 48.6 mL, (d) 57.1 mL, (e) 28,570 mL.

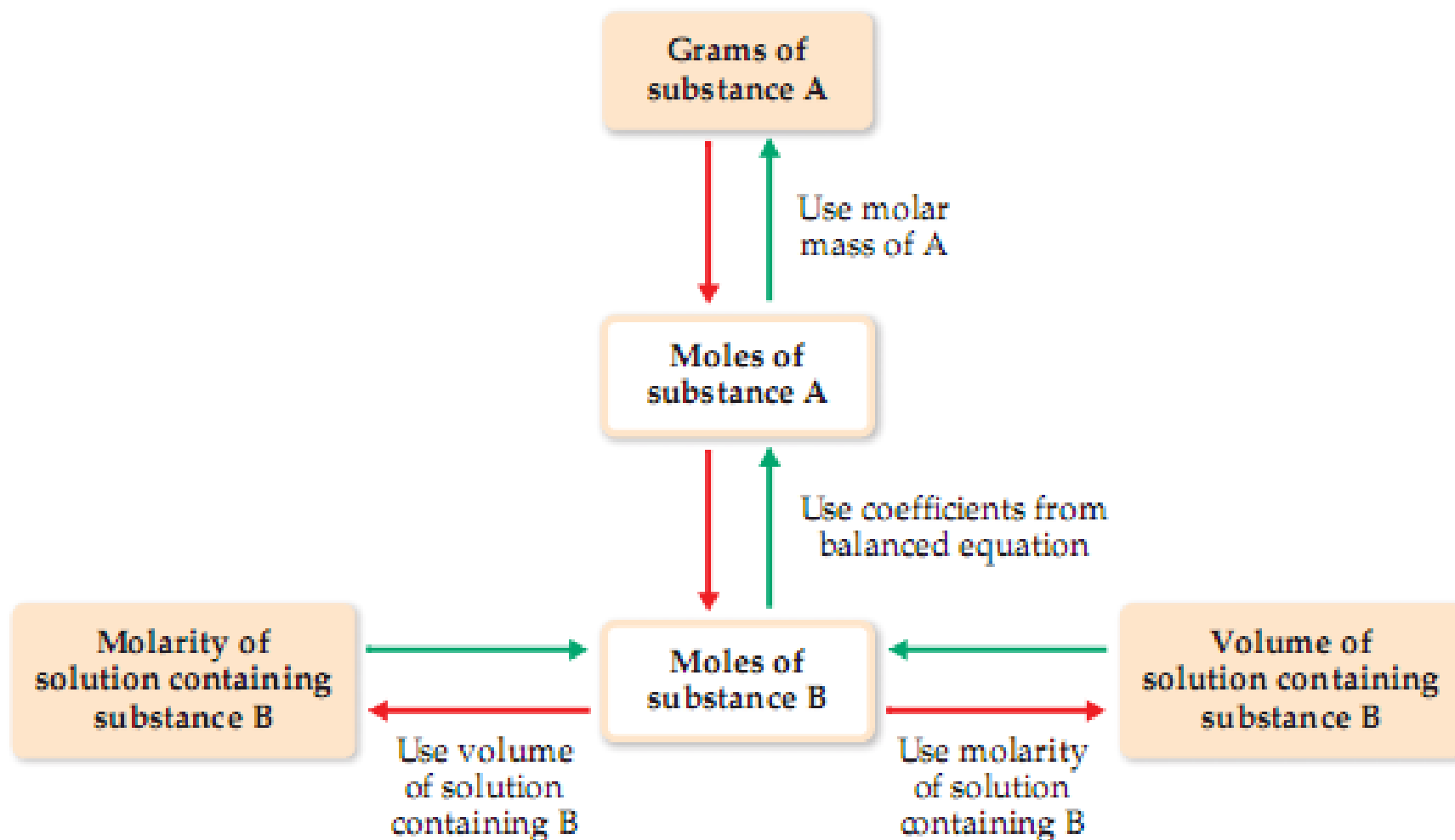
2.

(a) What volume of 2.50 M lead(II) nitrate solution contains 0.0500 mol of Pb^{2+} ?

(b) How many milliliters of 5.0 M $\text{K}_2\text{Cr}_2\text{O}_7$ solution must be diluted to prepare 250 mL of 0.10 M solution?

(c) If 10.0 mL of a 10.0 M stock solution of NaOH is diluted to 250 mL, what is the concentration of the resulting stock solution?

4-6 Solution Stoichiometry and Chemical Analysis



EXAMPLE

How many grams of $\text{Ca}(\text{OH})_2$ are needed to neutralize 25.0 mL of 0.100 M HNO_3 ?



$$V_{\text{HNO}_3} \times M_{\text{HNO}_3} \Rightarrow \text{mol HNO}_3 \Rightarrow \text{mol Ca}(\text{OH})_2 \Rightarrow \text{g Ca}(\text{OH})_2$$

$$\begin{aligned} \text{Moles HNO}_3 &= V_{\text{HNO}_3} \times M_{\text{HNO}_3} = (0.0250 \text{ L}) \left(\frac{0.100 \text{ mol HNO}_3}{\text{L}} \right) \\ &= 2.50 \times 10^{-3} \text{ mol HNO}_3 \end{aligned}$$

$$2 \text{ mol HNO}_3 \approx \text{mol Ca}(\text{OH})_2.$$

$$\begin{aligned} \text{Grams Ca}(\text{OH})_2 &= (2.50 \times 10^{-3} \text{ mol HNO}_3) \\ &\quad \times \left(\frac{1 \text{ mol Ca}(\text{OH})_2}{2 \text{ mol HNO}_3} \right) \left(\frac{74.1 \text{ g Ca}(\text{OH})_2}{1 \text{ mol Ca}(\text{OH})_2} \right) \\ &= 0.0926 \text{ g Ca}(\text{OH})_2 \end{aligned}$$

Titration.

◆ Titration

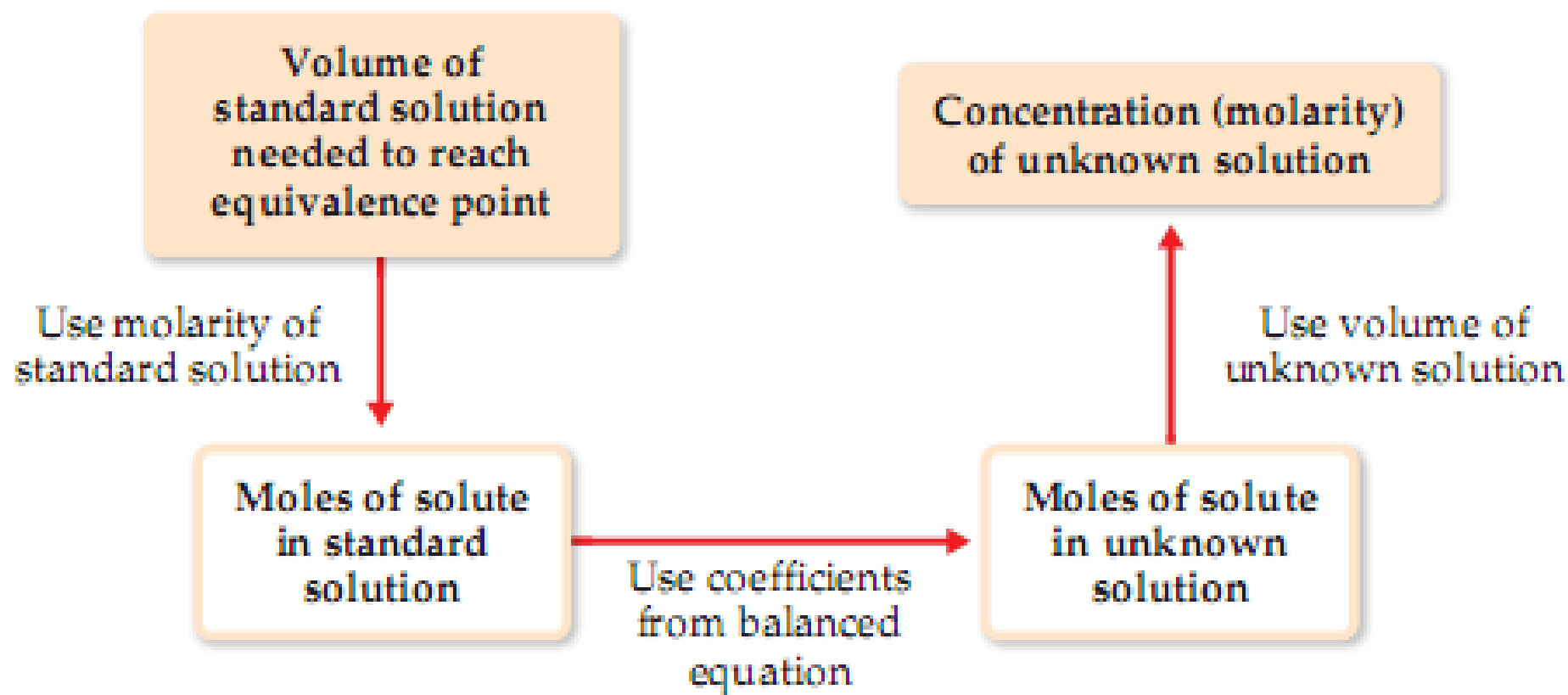
- Carefully controlled addition of one solution to another.

◆ Equivalence Point

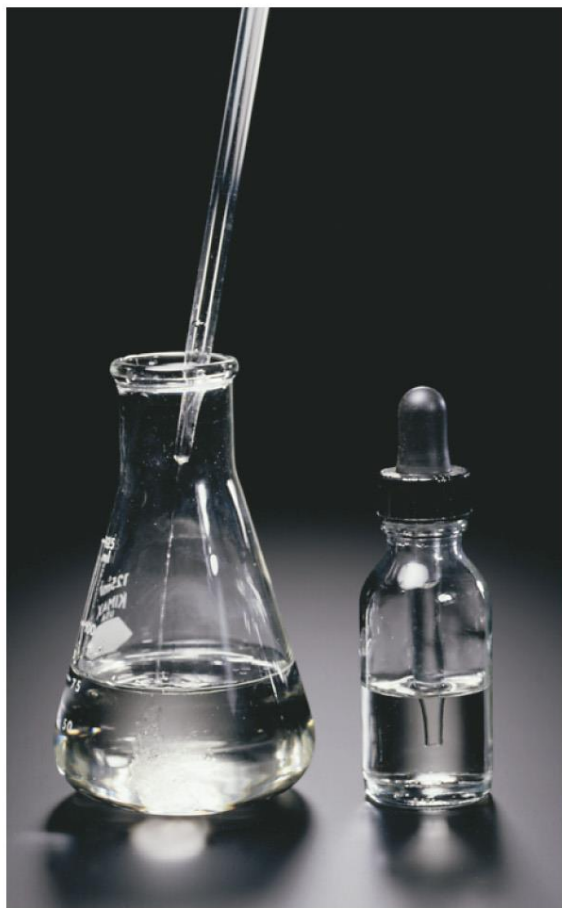
- Both reactants have reacted completely.

◆ Indicators

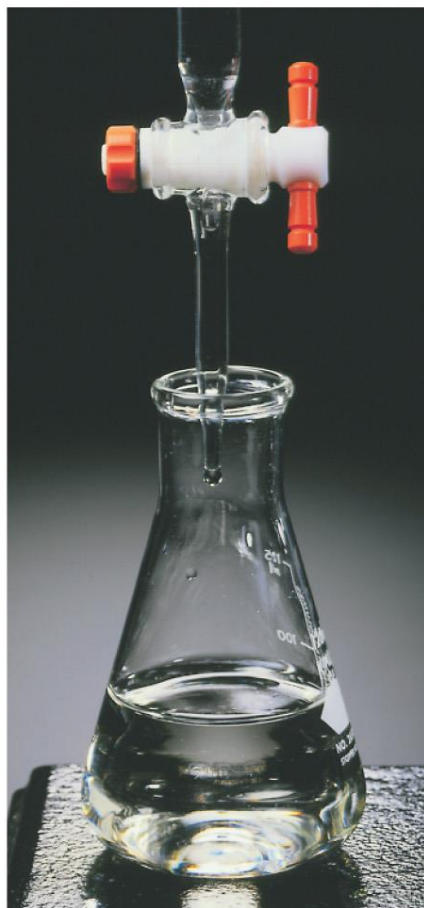
- Substances which change colour near an equivalence point.



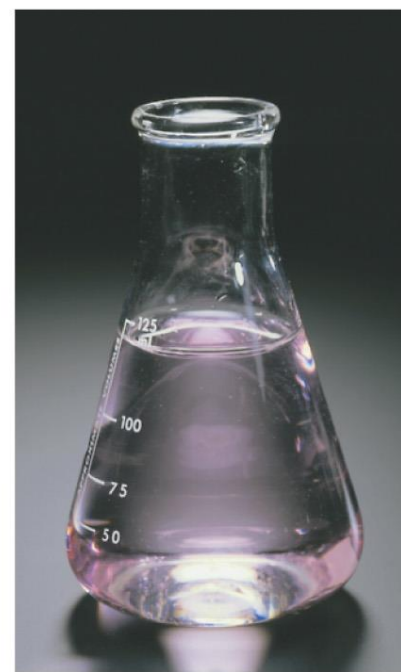
Indicators



5.0 mL $\text{CH}_3\text{CO}_2\text{H}$ A few drops
phenolphthalein



Add 0.1000 M NaOH

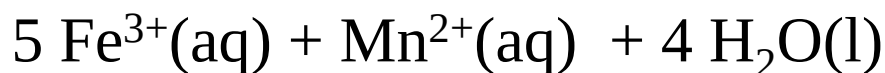
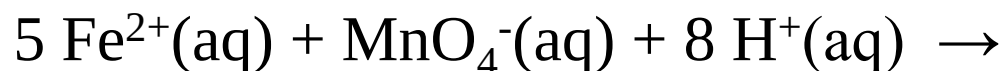


The “endpoint”
(close to the equivalence point)

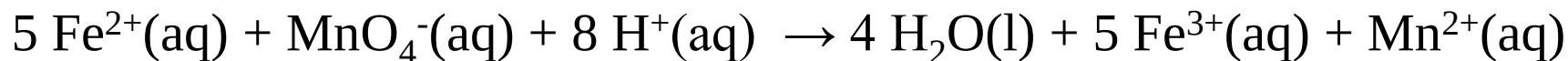
EXAMPLE

Standardizing a Solution for Use in Redox Titrations.

A piece of iron wire weighing 0.1568 g is converted to $\text{Fe}^{2+}(\text{aq})$ and requires 26.42 mL of a $\text{KMnO}_4(\text{aq})$ solution for its titration. What is the molarity of the $\text{KMnO}_4(\text{aq})$?



EXAMPLE



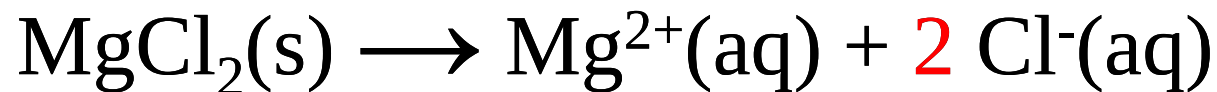
Determine KMnO_4 consumed in the reaction:

$$n_{\text{MnO}_4^{-}} = 0.1568 \text{ g Fe} \times \frac{1 \text{ mol Fe}}{55.847 \text{ g Fe}} \times \frac{1 \text{ mol Fe}^{2+}}{1 \text{ mol Fe}} \times \frac{1 \text{ mol MnO}_4^{-}}{5 \text{ mol Fe}^{2+}} \times \frac{1 \text{ mol KMnO}_4}{1 \text{ mol MnO}_4^{-}} = 5.615 \times 10^{-4} \text{ mol KMnO}_4$$

Determine the concentration:

$$c_{\text{KMnO}_4} = \frac{5.615 \times 10^{-4} \text{ mol KMnO}_4}{0.02624 \text{ L}} = 0.02140 \text{ M KMnO}_4$$

Relative Concentrations in Solution



In 0.0050 M MgCl_2 :

Stoichiometry is important.

$$[\text{Mg}^{2+}] = 0.0050 \text{ M} \quad [\text{Cl}^{-}] = 0.0100 \text{ M} \quad [\text{MgCl}_2] = 0 \text{ M}$$

EXAMPLE

Calculating Ion concentrations in a Solution of a Strong Electrolyte. What are the aluminum and sulfate ion concentrations in 0.0165 M $\text{Al}_2(\text{SO}_4)_3$?

Write a Balanced Chemical Equation:



Identify the Stoichiometric Factors :

$$\frac{2 \text{ mol Al}^{3+}}{1 \text{ mol Al}_2(\text{SO}_4)_3}$$

$$\frac{3 \text{ mol SO}_4^{2-}}{1 \text{ mol Al}_2(\text{SO}_4)_3}$$

EXAMPLE

Aluminum Concentration:

$$[\text{Al}] = \frac{0.0165 \text{ mol Al}_2(\text{SO}_4)_3}{1 \text{ L}} \times \frac{2 \text{ mol Al}^{3+}}{1 \text{ mol Al}_2(\text{SO}_4)_3} = 0.0330 \frac{\text{mol Al}^{3+}}{1 \text{ L}}$$

Sulfate Concentration:

$$[\text{SO}_4^{2-}] = \frac{0.0165 \text{ mol Al}_2(\text{SO}_4)_3}{1 \text{ L}} \times \frac{3 \text{ mol SO}_4^{2-}}{1 \text{ mol Al}_2(\text{SO}_4)_3} = 0.0495 \text{ M SO}_4^{2-}$$

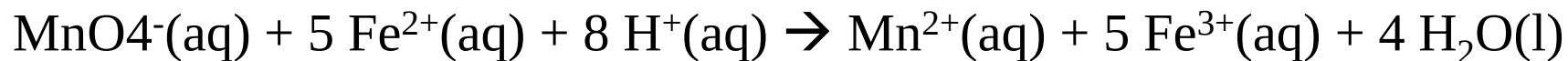
PRACTICE

1. A mysterious white powder is found at a crime scene. A simple chemical analysis concludes that the powder is a mixture of sugar and morphine ($\text{C}_{17}\text{H}_{19}\text{NO}_3$), a weak base similar to ammonia. The crime lab takes 10.00 mg of the mysterious white powder, dissolves it in 100.00 mL water, and titrates it to the equivalence point with 2.84 mL of a standard 0.0100 M HCl solution. What is the percentage of morphine in the white powder?

- (a) 8.10%,
- (b) 17.3%,
- (c) 32.6%,
- (d) 49.7%,
- (e) 81.0%.

PRACTICE

2. A sample of an iron ore is dissolved in acid, and the iron is converted to Fe^{2+} . The sample is then titrated with 47.20 mL of 0.02240 M MnO_4^- solution. The oxidation-reduction reaction that occurs during titration is



- (a) How many moles of MnO_4^- were added to the solution?
- (b) How many moles of Fe^{2+} were in the sample?
- (c) How many grams of iron were in the sample?
- (d) If the sample had a mass of 0.8890 g, what is the percentage of iron in the sample?

Homeworks

Exercises:

4.21

4.25

4.33

4.43

4.51

4.61

4.77

4.83

Due date: April 15th, 2020



You will have a 30 minute quiz next week

What are allowed?

- A pen
- A periodic table
- An A4-sized sheet of hand written notes
- A calculator

No other devices are allowed!